

Reproducible AI-Assisted Earthquake Early Warning Research for Western Seismic Regions

Method Development on a California FDSN Pilot with Transfer Path to
European Networks

29 July 2026

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Reproducible AI-Assisted Earthquake Early Warning Research for Western Seismic Regions

Method Development on a California FDSN Pilot with Transfer Path to European Networks

Final Research Proposal and Technical Report (Deliverable D-F1)

AMGCR Earthquake Research

29 July 2026

Certificate and programme statement

This document is submitted in partial fulfilment of research requirements for the Swiss certificate programme in Assessment and Management of Geological and Climate Related Risk (AMGCR), as defined in docs/PROJECT_CHARTER.md and docs/RESEARCH_DIRECTION.md.

The primary long-term research focus of the programme is Earthquake Early Warning methodology for European seismic networks. The United States (California) FDSN pilot documented herein is a completed methods laboratory, not the intended final deployment geography.

All quantitative results in the proposed work are drawn from existing repository artefacts and phase reports dated 2026-07-29. The proposed work does not re-run analysis scripts or introduce new scientific computations.

Evidence note: The repository does not record a student author name or CERG edition number. Programme affiliation text above is taken from reports/Research_Proposal_v1.md and docs/PROJECT_CHARTER.md only.

Executive summary

The AMGCR Earthquake Research project delivers a documented, script-driven workflow from FDSN earthquake catalogues and waveforms to machine-learning-ready features, aligned with European expectations for transparent natural-hazard research. Using ObsPy and public USGS (events) and EarthScope (waveforms) services, the pilot acquired eight vertical broadband (BHZ) MiniSEED recordings (magnitudes 4.05–4.87, 2024) and performed exploratory data analysis, STA/LTA-based onsite metrics, band-limited preprocessing, and extraction of an 8×18 feature matrix.

Main findings (completed pilot):

1. A reproducible end-to-end chain exists from manifest CSV through tables, JSON summaries, and figures under reports/.
2. Raw amplitudes in digital counts vary by orders of magnitude across events and stations (CI.ADO vs CI.USC), reflecting path, site, and gain rather than a homogeneous magnitude–amplitude relation at N=8.
3. P-wave trigger times from STA/LTA range 10.15–55.15 s after trace start; late picks on several records indicate trigger limitations without pre-event windows or analyst validation.
4. Preprocessing improves visual clarity and band-limited SNR proxies (mean peak-SNR 137 raw → 1684 filtered), but instrument response removal was not applied (0/8); amplitudes remain in counts, not physical ground-motion units.
5. Feature statistics and correlations are illustrative only at eight events; amplitude-derived features are highly collinear, while crest factor, entropies, and zero-crossing rate offer complementary shape information.

EarthESND (Joshi, Singh, and Raman, 2026) is retained as a complete reference implementation (88 passing tests on a full checkout) for future comparative machine-learning studies, not as the geographic or scientific endpoint of this certificate work.

European context: The same manifest-and-report pattern is proposed for ORFEUS/EIDA-class archives. No European waveform dataset is included in the completed pilot.

Submission artefacts: Raw MiniSEED, local acquisition logs, and generated PNG figures may be absent from Git clones (.gitignore); see Appendix C and docs/FINAL_DELIVERABLE_AUDIT.md.

Abstract

Earthquake Early Warning delivers rapid estimates of ground shaking ahead of destructive S-wave arrival, combining seismological data acquisition, onsite triggering, signal processing, and increasingly machine-learning-based magnitude and intensity prediction. European programmes require transparent, reproducible workflows that can be audited,

extended across national networks, and aligned with open data policies; however, many published pipelines remain tied to region-specific archives or proprietary tooling.

The proposed work presents the AMGCR Earthquake Research project: a European-focused certificate programme whose primary long-term target is Earthquake Early Warning-oriented analysis on European seismic networks, while California serves as the completed pilot implementation for method validation. Using ObsPy and FDSN services (USGS event metadata; EarthScope waveforms), eight MiniSEED recordings (magnitude 4.05–4.87, 2024) were acquired. Exploratory data analysis, STA/LTA-based signal characterisation, documented preprocessing, and extraction of an 8×18 feature matrix were executed. Results confirm a reproducible path from raw waveforms to machine-learning-ready descriptors, while exposing limitations that must be addressed before operational European deployment: origin-aligned download windows, absent instrument response correction in the pilot run, small sample size, and station-selection bias. Proposed future work includes scaling datasets, integrating ORFEUS/EIDA-class European archives, response-corrected amplitudes, and comparative evaluation of modern onsite artificial intelligence models on Western data. The reproducible workflow, scripts, and artefacts are available at <https://github.com/prasun-iitj/AMGCR-Earthquake-Early-Warning>.

Keywords: Earthquakes; Earthquake Early Warning Systems; Machine Learning; Deep Learning; ObsPy; reproducible research; feature engineering; Europe; California pilot; EarthScope; USGS.

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All paths relative to this file's directory (reports/). Full set: 24 PNGs per docs/FINAL_DELIVERABLE_AUDIT.md.

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1. Introduction

Damaging earthquakes remain a major risk in Europe and adjacent regions, from the Mediterranean belt to the Alps and anthropogenic seismicity associated with industrial activity. Earthquake Early Warning systems mitigate harm by issuing alerts seconds to tens of seconds before strong shaking arrives at a site, based on P-wave detection, rapid magnitude estimation, and ground-motion prediction. Public acceptance, regulatory frameworks, and civil-protection integration in Switzerland and across the European Union increasingly expect documented, open methodologies rather than black-box prototypes. Programme positioning, document scope, and document-control metadata are provided in Appendix F.

2. Literature review

Classical Earthquake Early Warning relies on onsite algorithms that detect P-wave onset via short-term average/long-term average (STA/LTA) ratio tests or similar triggers, then estimate magnitude and alert levels from early windows of acceleration or velocity (Allen and Kanamori, 2003; Hoshiba et al., 2008). European deployments emphasise standardised metadata, latency budgets, and uncertainty communication to end users.

ObsPy (Krischer et al., 2015) provides a standard interface for MiniSEED, FDSN web services, detrending, filtering, and instrument correction. In the proposed work, USGS supplies catalogues and EarthScope supplies waveforms (docs/IRIS_DATASET_REPORT.md); the same abstraction is intended for European endpoints (ORFEUS, EIDA, national nodes). Machine learning and deep learning approaches learn magnitude or intensity from early waveform segments but require large, diverse training sets and careful handling of site effects and instrument response.

Reproducible Earthquake Early Warning research demands versioned download parameters, manifests, processing configurations, and analysis scripts. This pattern is institutionalised through CSV manifests, JSON summaries, and phase reports. ORFEUS, EIDA, and member-state networks provide the intended long-term data plane; the California pilot functions as a methods prototype while the scientific narrative remains Europe-forward (docs/PROJECT_CHARTER.md).

3. Research gap

Despite growing interest in artificial-intelligence-assisted Earthquake Early Warning, certificate-level and early-stage research programmes in Europe still lack widely

accessible, end-to-end workflows that connect open FDSN archives to auditable feature matrices suitable for method development. Many published pipelines remain region-specific or proprietary, and small-N pilots, while valuable for prototyping, cannot support operational claims without scaled catalogues, validated arrival picks, and consistent metadata handling across networks. The proposed work responds to this gap by delivering a documented, script-driven California pilot that makes limitations explicit and defines a transferable pattern for future ORFEUS/EIDA integration (docs/RESEARCH_DIRECTION.md; docs/PROJECT_CHARTER.md).

4. Objectives

The proposed work addresses five objectives aligned with completed pilot artefacts and stated future extensions:

- O1. Develop a reproducible, artificial-intelligence-ready Earthquake Early Warning research workflow aimed ultimately at European seismic networks, documented for a Swiss and European research audience. (*Programme objective; California pilot workflow completed.*)
 - O2. Acquire real earthquake waveforms through ObsPy and FDSN services using a California IRIS/EarthScope pilot (eight events, vertical broadband BHZ channel). (*Completed.*)
 - O3. Characterise the pilot dataset through exploratory data analysis, onsite-style STA/LTA signal metrics, and a documented preprocessing chain toward machine-learning inputs. (*Completed.*)
 - O4. Extract a multi-domain feature matrix (8×18) suitable for exploratory machine-learning and deep-learning experiments on Western data. (*Completed.*)
 - O5. Interpret pilot results, document limitations and threats to validity, and define a European transfer pathway aligned with ORFEUS/EIDA-class archives. (*Completed for California pilot; European acquisition proposed.*)
-

5. Dataset

The completed pilot dataset comprises eight California earthquake recordings acquired through public FDSN services: event metadata from USGS and waveforms from EarthScope (formerly IRIS DMC). California was selected as the pilot region because programme requirements in docs/PROJECT_CHARTER.md (§4) mandate ObsPy-based download of real waveforms from IRIS/EarthScope for method development, while the programme’s long-term focus remains European seismic networks. California offers a well-documented, publicly accessible FDSN infrastructure, a seismically active test bed in the M 4–5 band suitable for pilot-scale workflow validation, and a practical laboratory for exercising acquisition, STA/LTA analysis, preprocessing, and feature engineering before

extending the same manifest-and-report pattern to European endpoints. The pilot is a methods laboratory, not the intended final deployment geography (docs/RESEARCH_DIRECTION.md).

5.1 Completed California pilot

Table 1. Pilot dataset attributes (sources: docs/IRIS_DATASET_REPORT.md, docs/EDA_REPORT.md, reports/eda/dataset_summary.json)

Attribute	Value
Region	California bounding box, 2024
Event provider	USGS FDSN Event
Waveform provider	EarthScope (Client("EARTHSCOPE"))
Events in manifest	8
Magnitude range	4.05 – 4.87 (mw, ml)
Origin times (UTC)	2024-05-01 – 2024-10-25
Channel	BHZ
Sampling rate	40 Hz
Window	300 s from catalog origin time
Stations (manifest)	CI.ADO (6 events), CI.USC (2 events)
Manifest path	data/manifests/iris_california_pilot_events.csv
Raw waveforms (local)	data/raw/iris/

Legacy files: EDA recorded twelve MiniSEED files on disk at run time; four lie outside the current manifest (reports/eda/mseed_file_inspection.csv). All completed analyses use eight manifest-linked records only.

5.2 FDSN acquisition parameters (summary)

Table 2. FDSN query parameters (from docs/IRIS_DATASET_REPORT.md)

Parameter	Event search (USGS)	Waveform download (EarthScope)
Time window	2024-01-01 – 2024-12-31	300 s from origin
Latitude	32.5° – 42.0° N	—
Longitude	–124.5° – –114.0° E	—
Magnitude	4.0 – 7.5	—
Networks (candidates)	—	CI, NC, BK
Channel	—	BHZ (may try HHZ/EHZ)
Stations per event	—	1 (first successful candidate)
Catalog matches (query)	41	—

Parameter	Event search (USGS)	Waveform download (EarthScope)
Events retained	8	8 waveforms

Station selection uses a fixed distance-ranked candidate list; this is not uniform network sampling.

5.3 European dataset (proposed)

No European waveforms are included. Future work is expected to mirror manifest and report patterns for EIDA/ORFEUS (docs/PROJECT_CHARTER.md, docs/ROADMAP.md).

5.4 Published tables and matrices

- reports/tables/event_summary.csv, station_summary.csv
- reports/tables/signal_analysis_per_event.csv
- reports/tables/preprocessing_quality_metrics.csv
- reports/features/feature_matrix.csv,
reports/tables/feature_correlation_matrix.csv

6. Methodology

6.1 Acquisition (completed)

Script: scripts/download/download_iris_california_pilot.py. Events from USGS; waveforms from EarthScope dataset. Parameters and results: docs/IRIS_DATASET_REPORT.md, logs/iris_california_pilot_summary.json (local, gitignored).

6.2 Exploratory data analysis (Phase A.1)

Script: scripts/analysis/run_california_pilot_eda.py. ObsPy inspection of all .mseed under data/raw/iris/; cohort summaries for manifest events only (docs/EDA_REPORT.md).

6.3 Signal analysis (Phase A.2)

Script: scripts/analysis/run_california_signal_analysis.py. Per trace: peak/RMS amplitudes, noise RMS from 0–5 s proxy, peak SNR, STA/LTA P pick (STA 0.5 s, LTA 10 s, threshold 2.5, ObsPy classic_sta_lta), signal duration (docs/SIGNAL_ANALYSIS_REPORT.md).

6.4 Preprocessing (Phase B.1)

Script: `scripts/analysis/run_california_preprocessing.py`. Frozen config: `reports/preprocessing/preprocessing_config.json`.

Table 3. Preprocessing parameters

Step	Setting
Demean	Mean removal
Detrend	Linear
Taper	Hann, 5%
Band-pass	0.1–15 Hz, four corners, zero-phase
Instrument response	EarthScope level=response → velocity (attempted)
Normalisation	Peak absolute amplitude

Outcome: Response removal 0/8 applied (skipped); filtered traces remain in counts (docs/PREPROCESSING_REPORT.md §4). Stages: `reports/preprocessing/<short_id>_stages.npz`.

6.5 Feature engineering (Phase B.2)

Script: `scripts/analysis/run_california_feature_engineering.py`. 18 features from filtered traces; P times from Phase A.2 metrics JSON (docs/FEATURE_ENGINEERING_REPORT.md).

Table 7. Feature domains (18 features)

Domain	Count	Examples (repository names)
Time-domain	9	<code>peak_amplitude</code> , <code>rms</code> , <code>crest_factor</code> , <code>zero_crossing_rate_hz</code> , <code>signal_entropy</code> , ...
Frequency-domain	5	<code>dominant_frequency_hz</code> , <code>spectral_centroid_hz</code> , <code>spectral_bandwidth_hz</code> , <code>spectral_rolloff_95_hz</code> , <code>spectral_entropy</code>
Earthquake-related	4	<code>p_arrival_time_s</code> , <code>signal_duration_s</code> , <code>pgm_proxy_counts</code> , <code>arias_intensity_counts2_s_proxy</code>

FFT spectra are figure-only (not matrix columns). PGM and Arias entries are counts-based proxies, not physical PGA or standard Arias intensity.

6.6 EarthESND reference (not primary experiment)

Full EarthESND-aligned implementation in `src/` with 88 passing tests on a complete checkout. K-NET scientific reproduction is not started. California pilot results did not use EarthESND modules (docs/RESULTS_AND_DISCUSSION.md §1).

7. California pilot workflow and execution

[Completed] USGS + EarthScope download → MiniSEED + CSV manifest
[Completed] Phase A.1 EDA → reports/eda/, four summary figures
[Completed] Phase A.2 Signal analysis → STA/LTA, SNR, eight four-panel figures
[Completed] Phase B.1 Preprocessing → stage NPZ, eight comparison figures
[Completed] Phase B.2 Features → 8×18 matrix, four feature figures
[Completed] Phase C Interpretation → docs/RESULTS_AND_DISCUSSION.md
[Completed] Deliverable 1 → Research_Proposal_v1.md
[Completed] D-S1/D-S2 → Reproducibility & data availability statements
[Completed] D-F1 → final research report (this document)
[Proposed] European acquisition, response cache, larger N, ML, real-time

Reproduction commands: Appendix B. The EarthESND reference track remains optional (docs/PAPER_REPRODUCTION.md).

8. Results

Primary synthesis: docs/RESULTS_AND_DISCUSSION.md. No new numbers below.

8.1 Exploratory data analysis

Eight events occupy a narrow M 4–5 band; CI.ADO supplies six of eight traces. Figures 1–4 summarise raw example, magnitude distribution, temporal coverage, and station usage.

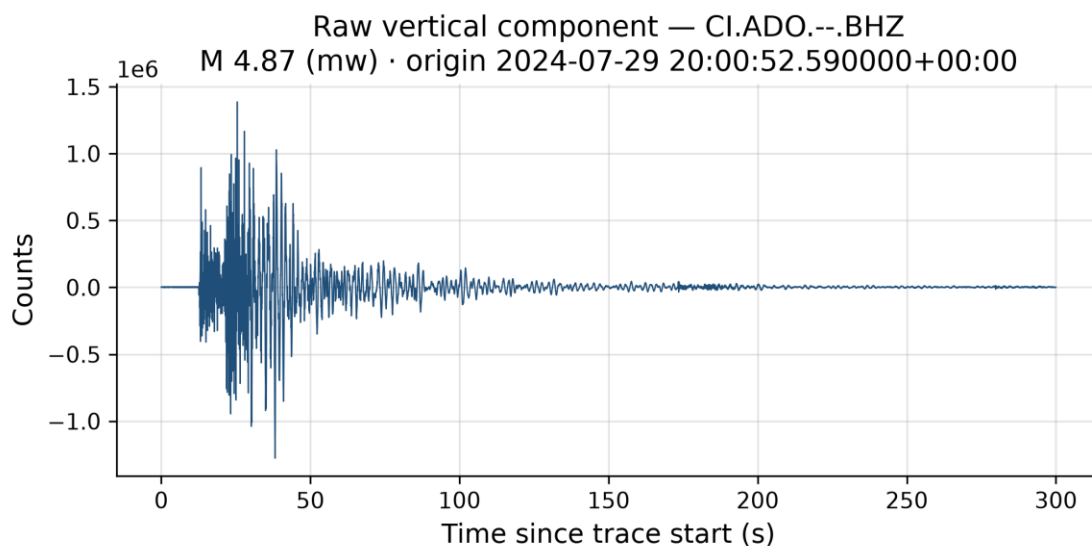


Figure 1: Raw BHZ record (highest pilot magnitude M 4.87, CI.ADO); amplitudes in digital counts, 300 s window from trace start.

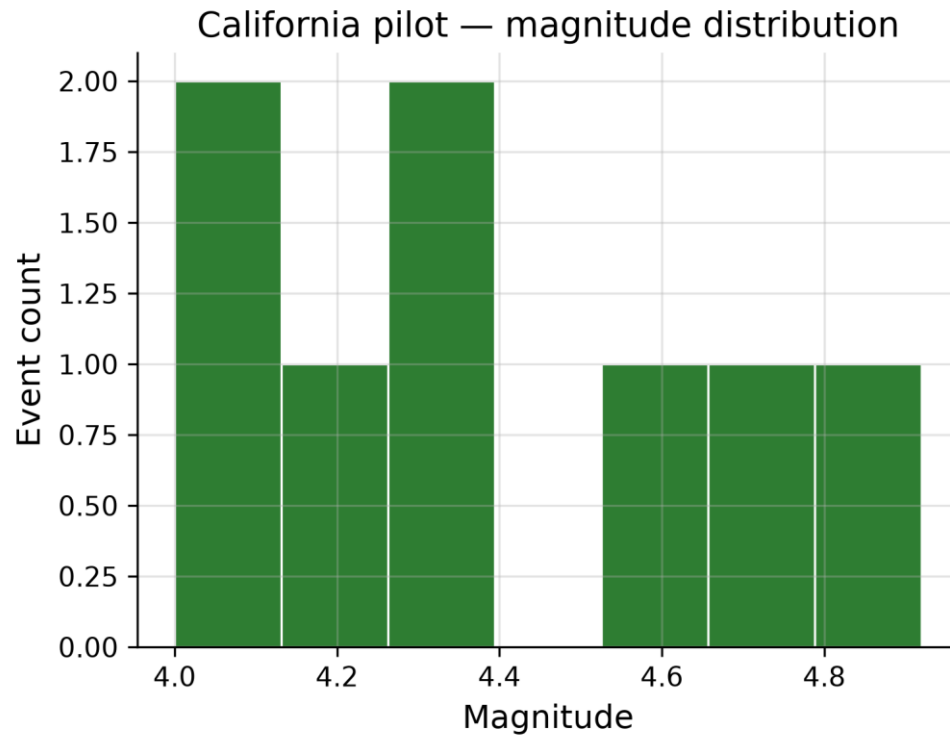


Figure 2: Magnitude distribution for eight manifest events.

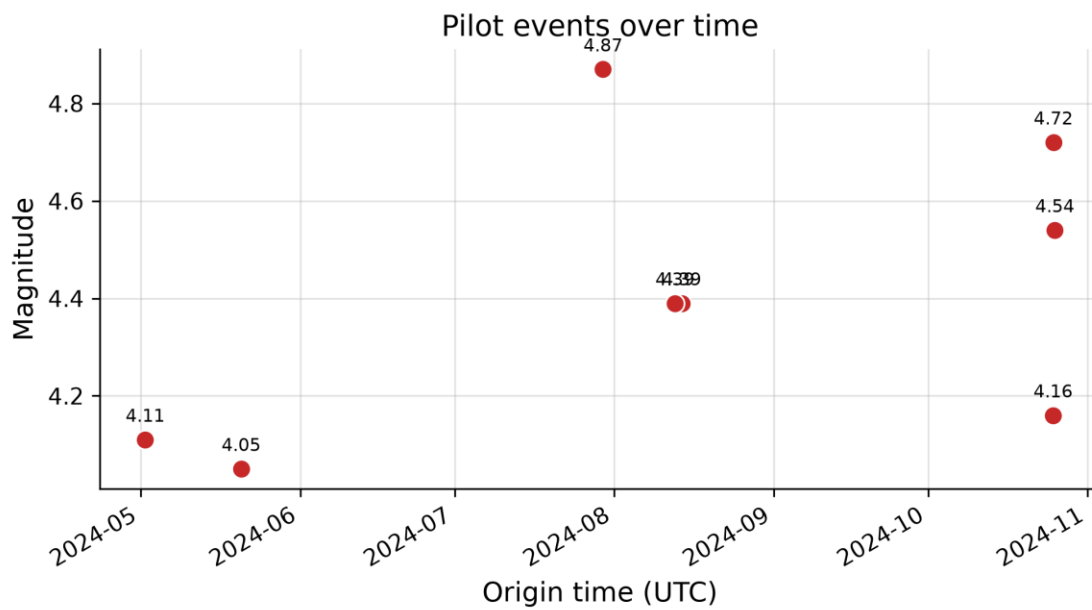


Figure 3: Origin time versus magnitude for manifest events.

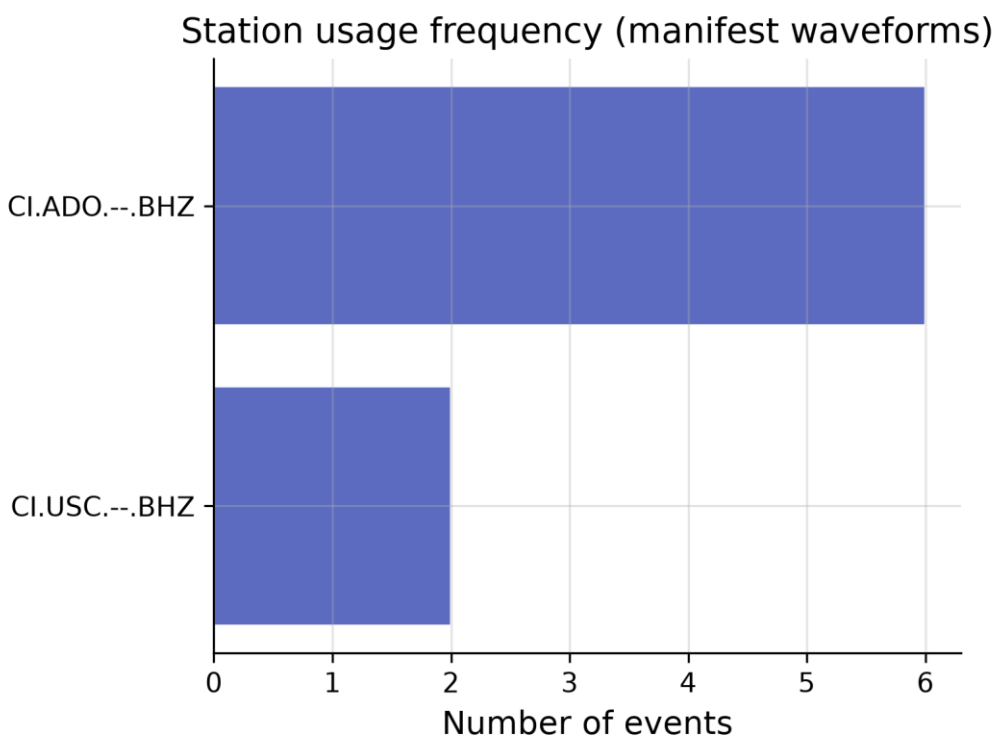


Figure 4: Station usage frequency (CI.ADO vs CI.USC).

Interpretation (docs/RESULTS_AND_DISCUSSION.md §2.3): The histogram reflects query cutoffs, not regional completeness. Temporal gaps reflect download success. ADO dominance is structural bias from the acquisition script.

8.2 Signal analysis and waveform behaviour

Peak raw amplitudes span approximately 1.7×10^4 to 3.5×10^6 counts (reports/signal_analysis/signal_analysis_summary.json).

Table 4. Aggregate signal metrics (eight events)

Quantity	Value
P pick (s from trace start)	10.15 – 55.15 (mean 31.3)
SNR (peak), raw	6.3 – 778 (median 49.9)
Signal duration (post-P proxy)	0 – 289.5 s (mean 183.2)

Table 5. Station-level signal contrasts (signal_analysis_summary.json)

Station	n	Mean SNR (peak)	Mean P pick (s)	Mean peak (counts)
CI.ADO	6	169	30.5	2.98×10^5
CI.USC	2	42	33.6	1.83×10^6

Figure 5 shows ci40675215 (M 4.87, high SNR, early pick ~10.2 s). Figure 6 shows ci40699207 at CI. USC (SNR ~6, duration 0 s), illustrating noise-window contamination when downloads start at origin time.

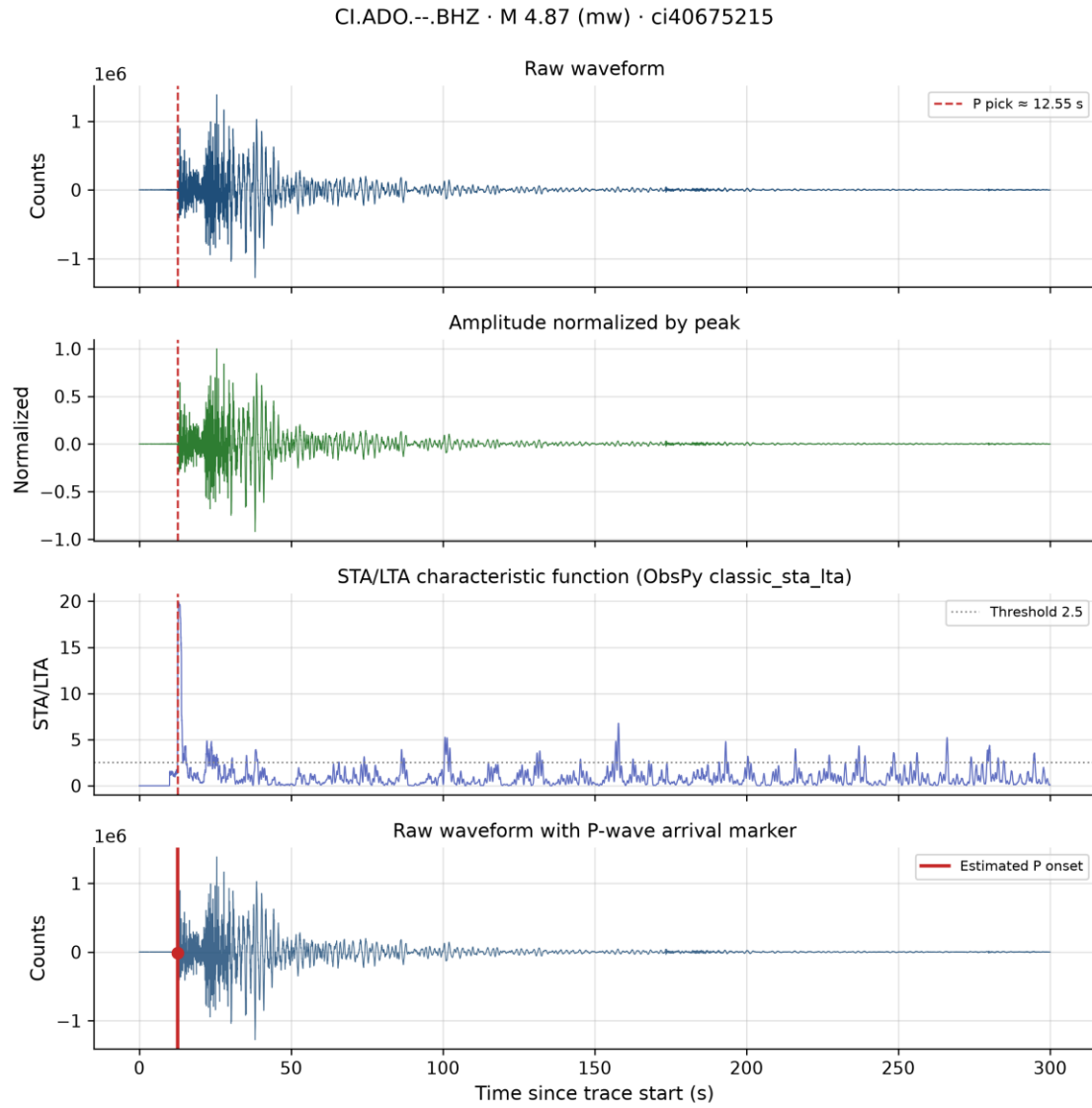


Figure 5: Signal analysis panels for ci40675215 (raw, normalised, STA/LTA, P marker).

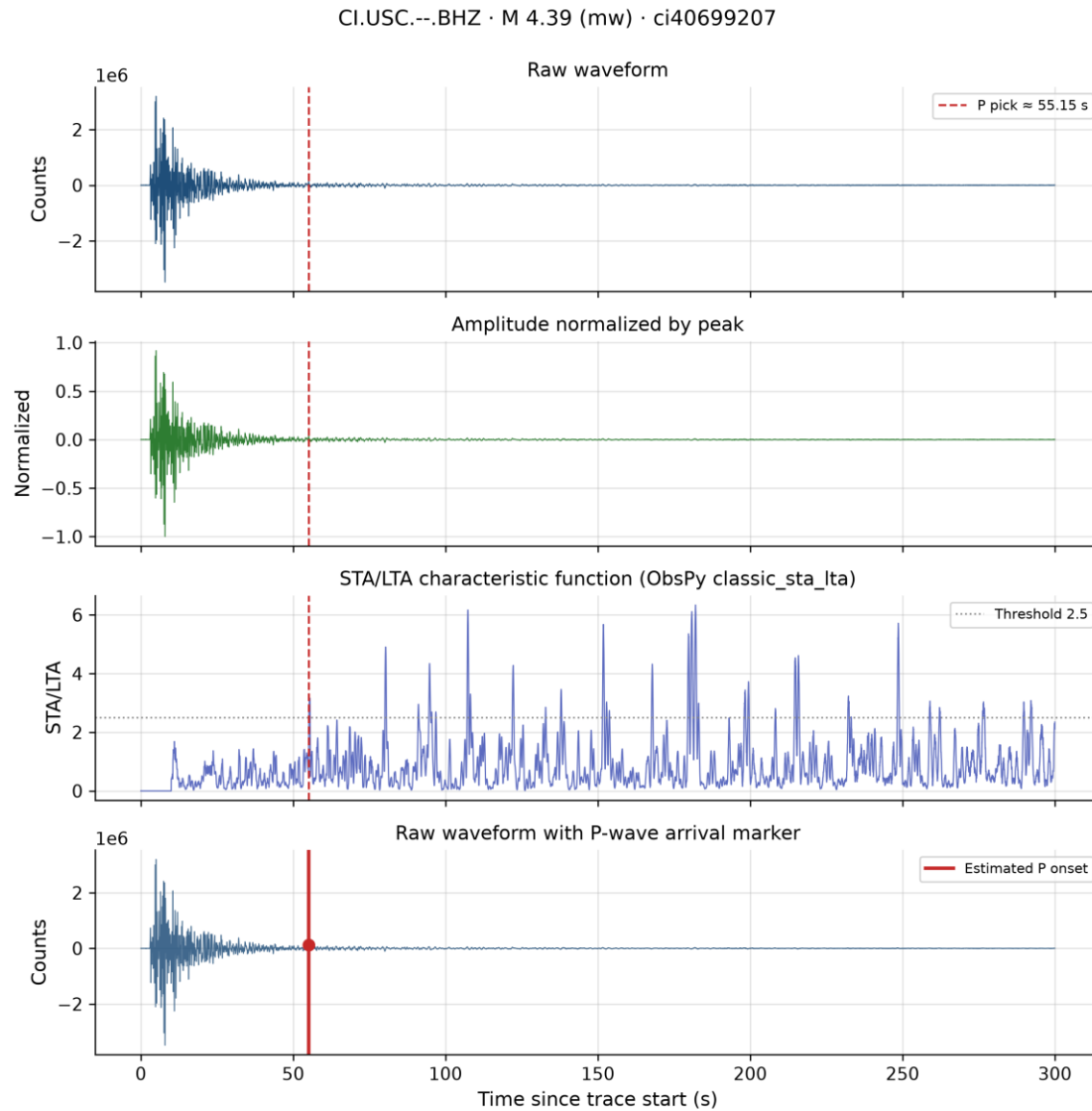


Figure 6: Signal analysis panels for ci40699207 (low peak-SNR case at USC).

8.3 Preprocessing

Table 6. Preprocessing SNR aggregates

Stage	Mean SNR (peak)
Raw	137
Filtered (pre-normalisation)	1684

Filtered/raw SNR ratio range 2.5–27.6 (mean 17.1). Figures 7–8 show four-panel progressions (raw → detrended → filtered → normalised) for the same events as Figures 5–6.

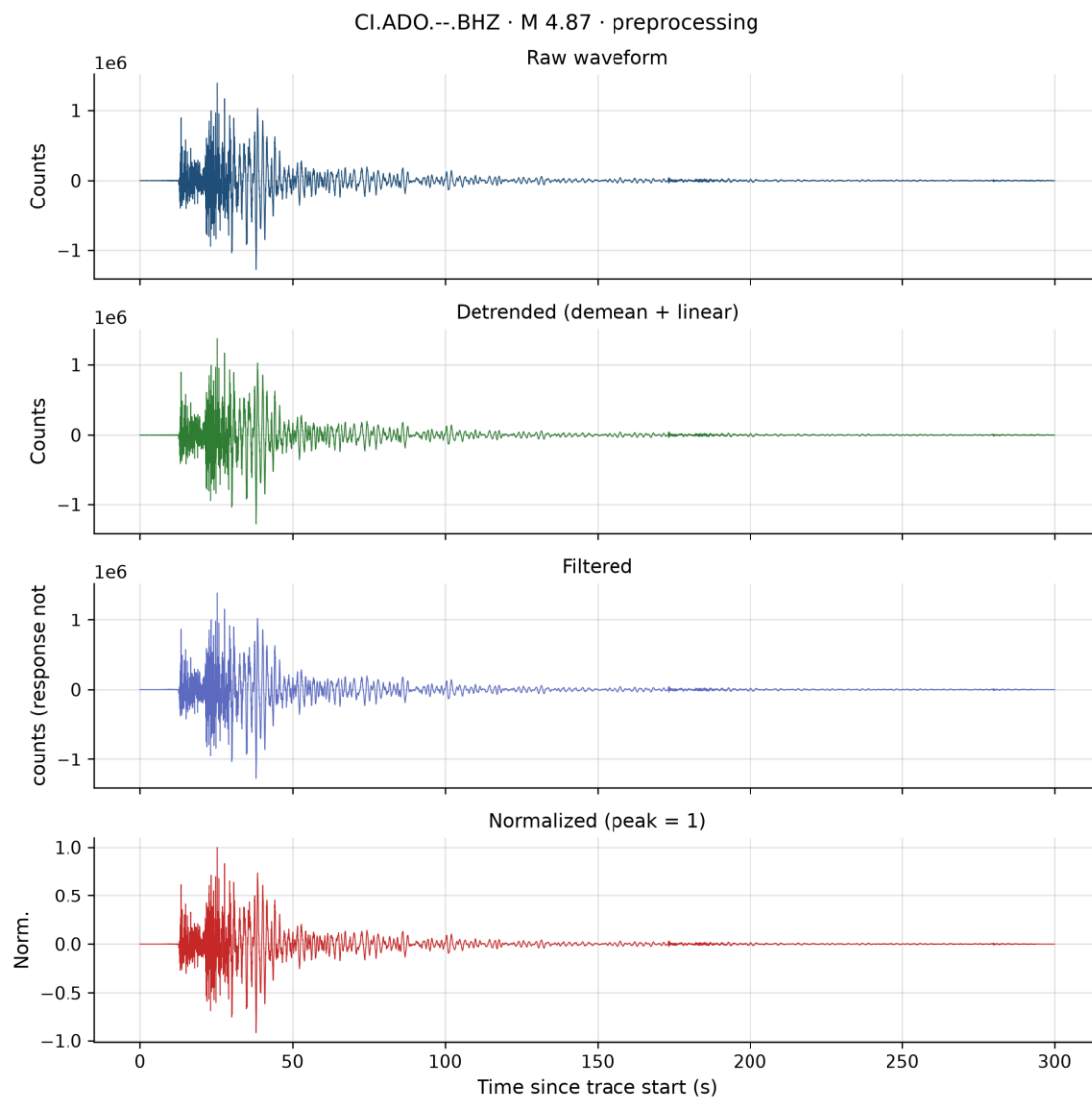


Figure 7: Preprocessing stages for ci40675215.

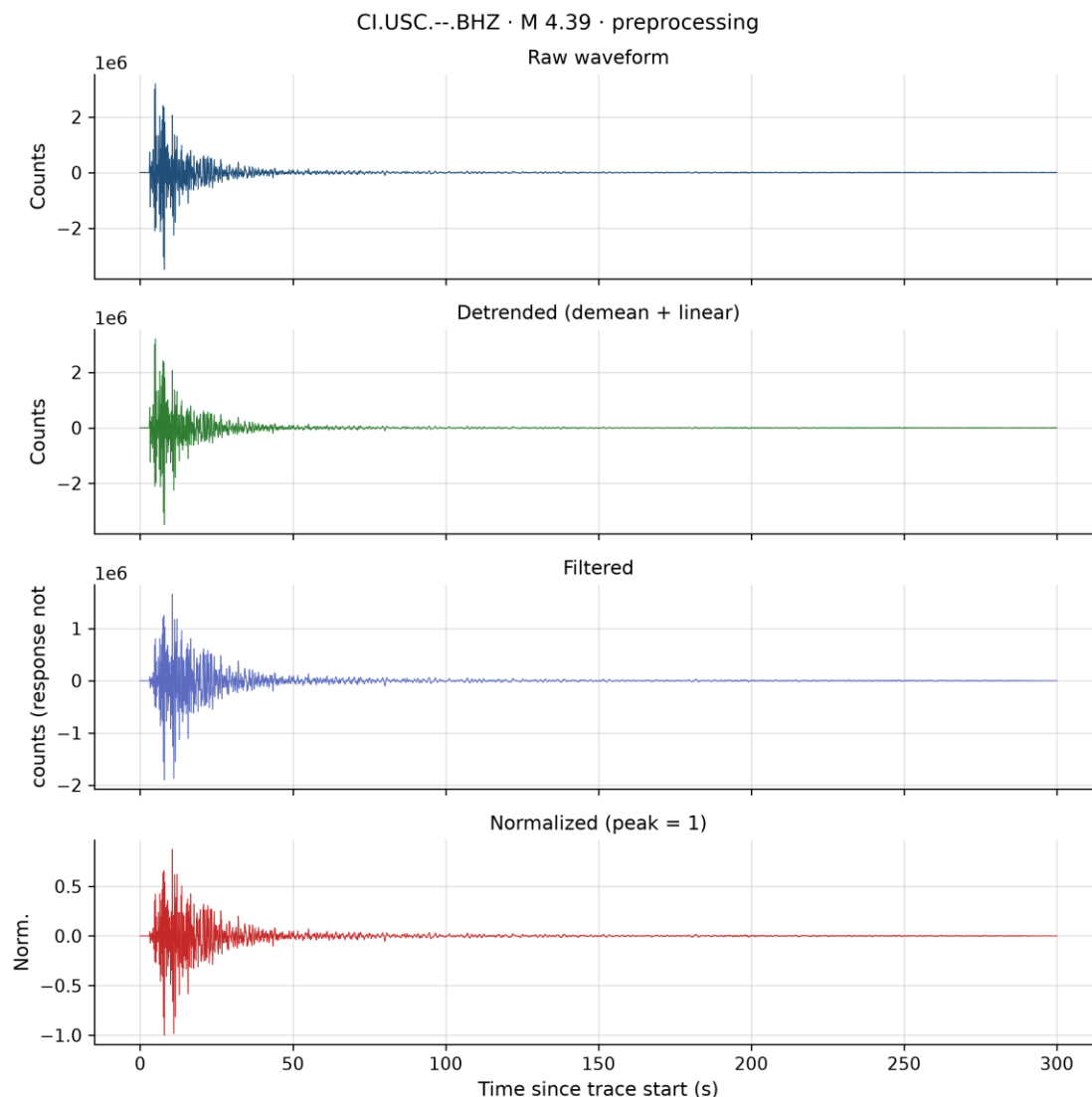


Figure 8: Preprocessing stages for ci40699207.

Interpretation: Visual stabilisation supports machine-learning-oriented comparison; without response correction, filtered counts are not physical velocity or acceleration for magnitude estimation.

8.4 Feature engineering

The 8×18 matrix is stored in `reports/features/feature_matrix.csv`. Crest factor cohort mean ≈ 11.4 ; dominant frequency and spectral centroid cluster near $\sim 2.3\text{--}2.6$ Hz (`reports/features/feature_summary.json`), consistent with the 0.1–15 Hz band-pass.

Strong collinearity links amplitude-derived features; crest factor, signal entropy, spectral entropy, and `zero_crossing_rate_hz` provide partially independent shape information at $N=8$ only.

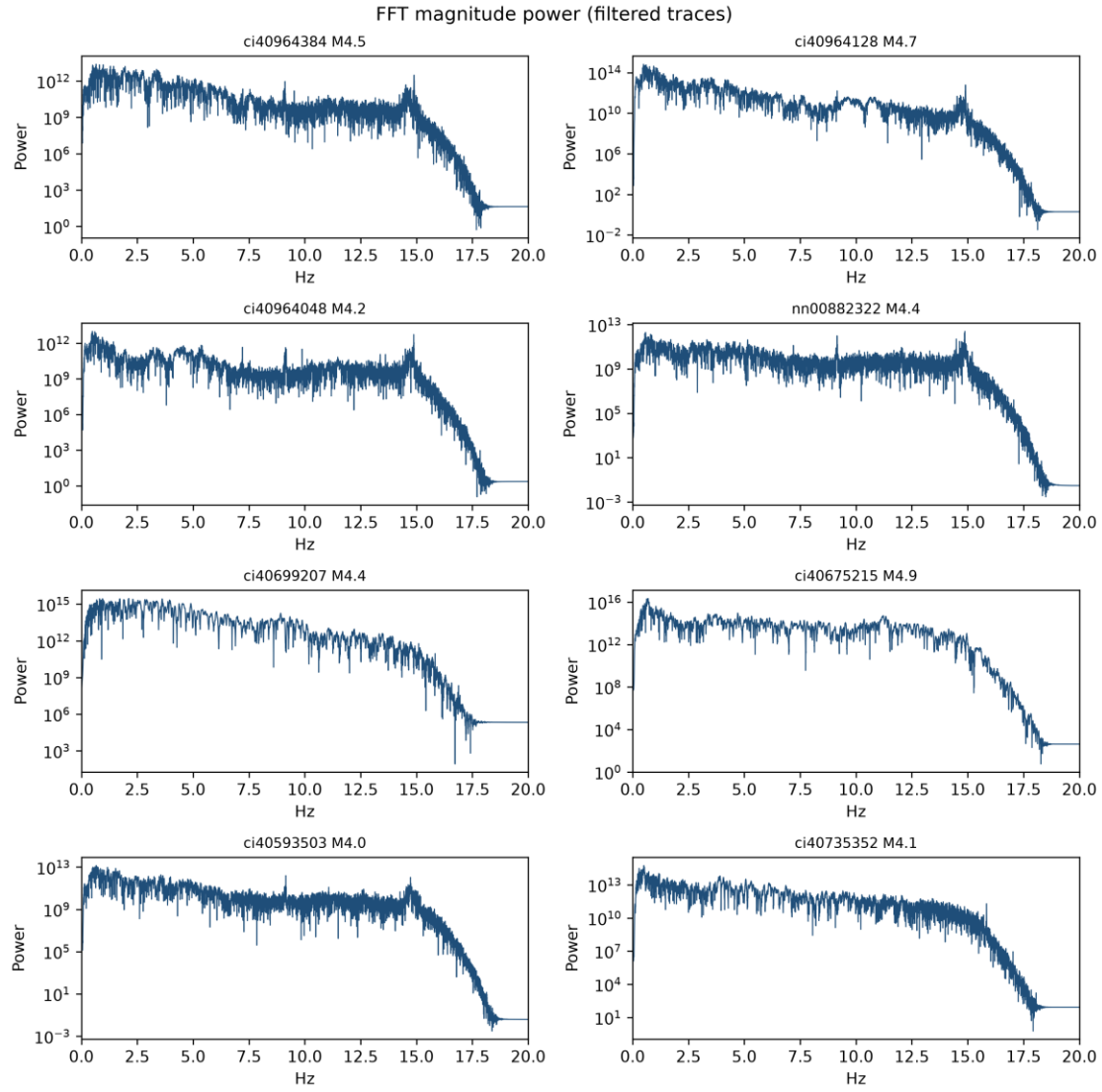


Figure 9: Per-event FFT power spectra (filtered traces).

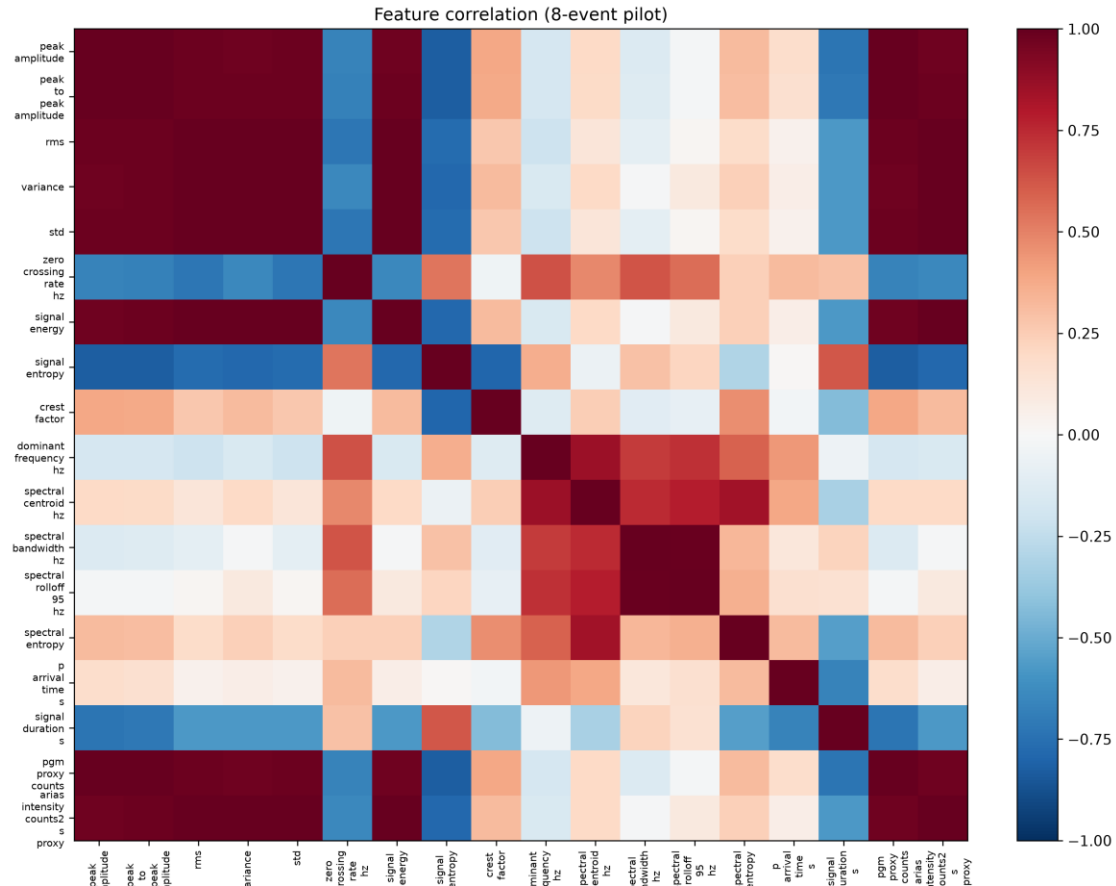


Figure 10: Pearson correlation heatmap (18 features, 8 samples).

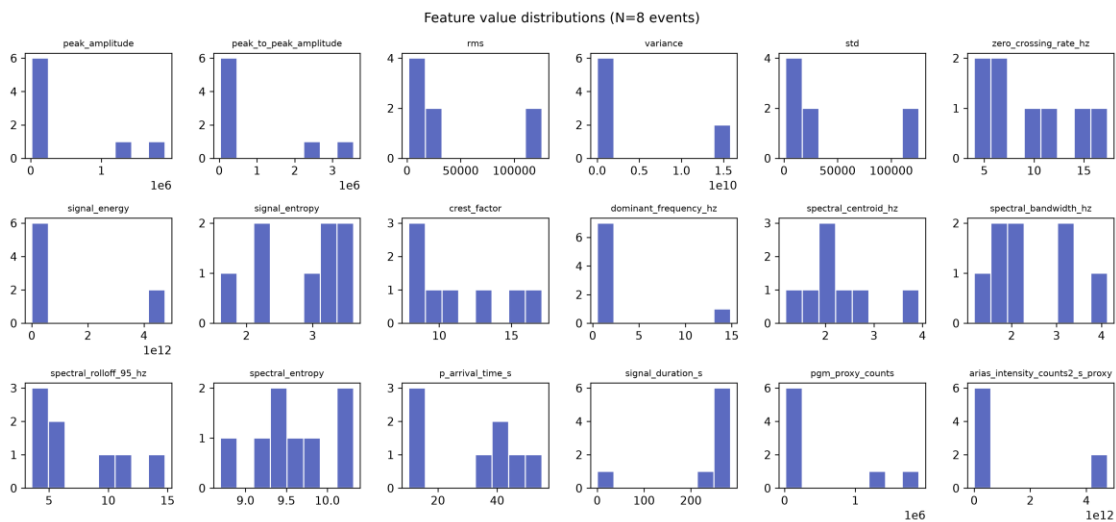


Figure 11: Per-feature distributions.

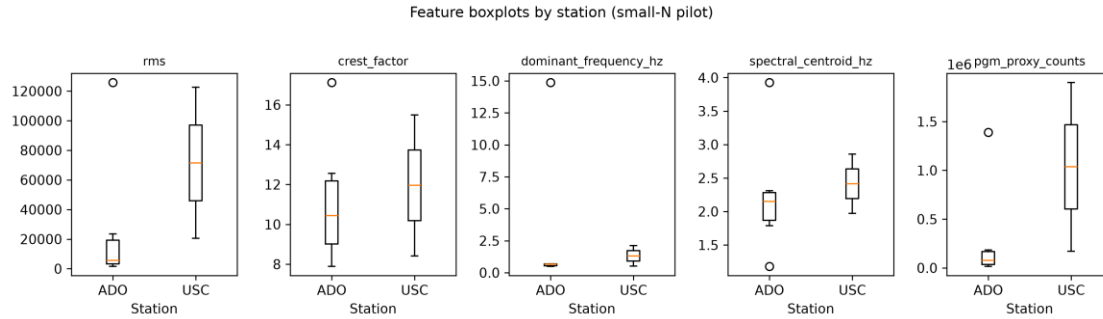


Figure 12: Selected features by station (ADO vs USC).

9. Discussion

The pilot demonstrates a script-driven, documented path from FDSN services to tabular machine-learning inputs and normalised NPZ stages, satisfying certificate requirements for ObsPy, real data, graphs, and tables (docs/RESULTS_AND_DISCUSSION.md §8). STA/LTA panels connect classroom Earthquake Early Warning theory to P-time features, but late picks and origin-aligned noise show that trigger parameters cannot be tuned on this dataset alone without pre-event windows and validation. The feature matrix is suitable for pilot baselines (for example, linear models or random forests) only in an exploratory sense; no machine-learning training or hold-out evaluation was performed in the completed pilot, and no predictive performance metrics are reported in repository artefacts. Reference code in src/ enables future architecture comparison on Western tensors with an explicit deviation log from Japan-centric paper tables, but it does not validate EarthESND on California data in the completed work.

10. Limitations and threats to validity

10.1 Pilot limitations (California)

1. N = 8 — no reliable generalisation or correlation inference.
2. Single component (BHZ).
3. Origin-aligned windows — biased noise and SNR.
4. No response removal (0/8).
5. Station selection bias (ADO-heavy; USC n=2).
6. Mixed magnitude types (mw, m1).
7. Four legacy MiniSEED files on disk outside manifest.
8. Unvalidated P picks.
9. Not comparable to EarthESND K-NET metrics.

(Source: docs/RESULTS_AND_DISCUSSION.md §6.)

10.2 Threats to validity

Table 8. Threats to validity and proposed mitigations (from docs/RESULTS_AND_DISCUSSION.md §7)

Threat	Description	Mitigation (proposed)
Construct validity	Counts used where physical motion required	StationXML cache; response removal
Internal validity	SNR inflated after filtering with fixed noise window	Pre-origin download windows
External validity	California CI $\neq$ European networks	EIDA/ORFEUS pilot with same scripts
Statistical conclusion validity	N=8	Larger catalogues; bootstrap CIs
Selection bias	First-eight successful downloads	Stratified sampling
Confirmation bias	Fixed STA/LTA threshold	Threshold sweeps; ML picks

11. European context and transfer

Europe is the primary programme focus; California validates methods, not deployment geography (docs/PROJECT_CHARTER.md).

Why the pilot supports European goals (docs/RESULTS_AND_DISCUSSION.md §9):

- Demonstrates ObsPy/FDSN reproducibility.
- Surfaces pitfalls (origin-aligned windows, response metadata, station bias) relevant to ORFEUS/EIDA integration.
- Delivers figures and tables for methodology chapters while European catalogues are scoped.

Transfer plan (proposed): Reuse manifest CSV structure, phase report templates, and scripts/ layout with European endpoint configuration; document licensing, magnitude conventions, and channel choices per network.

Expected challenges (proposed): Fragmented access policies, lower event rates in some regions, heterogeneous station density, accelerometer vs broadband mix, regulatory metadata (docs/RESULTS_AND_DISCUSSION.md §9.3).

Expected benefits (proposed): Cross-region feature comparison, open artefact trails, alignment with Swiss/European Earthquake Early Warning research narratives.

12. Future work and expected outcomes

12.1 Already achieved

- Documented California FDSN acquisition and full analysis chain.
- Phase reports, reports/ artefacts, Research Proposal v1, D-S1/D-S2, and the D-F1 final report.
- EarthESND reference codebase for comparative studies (not primary objective).

12.2 Proposed extensions

All items below denote proposed future work (docs/ROADMAP.md; docs/RESULTS_AND_DISCUSSION.md §10).

Direction	Objective
Larger catalogues	50–200+ events; stratified splits
European FDSN pilot	EIDA/ORFEUS manifests and reports
Instrumentation	Response correction; 3-component data; pre-event noise
AI models	Tabular and waveform baselines; EarthESND-inspired comparisons with deviation log
Real-time Earthquake Early Warning	SeedLink simulation; latency budgets
Formal publication	European transfer with uncertainty quantification

(Source: docs/RESULTS_AND_DISCUSSION.md §10, reports/Research_Proposal_v1.md §11.)

13. Conclusion

1. A reproducible chain from FDSN acquisition to an 8×18 feature matrix is established for the California pilot.
2. Raw waveforms exhibit strong station and path effects and heterogeneous STA/LTA pick times; CI.ADO dominates sampling.
3. Preprocessing improves interpretability and band-limited SNR proxies, but response removal failed for all eight traces; physical amplitude Earthquake Early Warning is not yet supported.

4. Features combine time, frequency, and pick-based descriptors; correlations show redundancy among amplitude metrics and value in shape statistics at pilot scale.
 5. Validity threats must be addressed before operational or publication-grade Earthquake Early Warning claims.
 6. California proves the method; Europe is the intended context for scaled deployment with network-specific metadata discipline.
-

14. References

Allen, R. M., and Kanamori, H. (2003). The potential for earthquake early warning in southern California. *Science*, 300(5620), 786–788. <https://doi.org/10.1126/science.1080532>

Hoshiba, M., Iwakiri, K., Hayashimoto, N., Shimoyama, T., Hirano, K., Yamada, Y., Ishigaki, Y., and Kikuta, H. (2008). Outline of the 2007–2008 earthquake early warning experiments in Japan. *Earth, Planets and Space*, 60, 123–129. <https://doi.org/10.1186/BF03352879>

Joshi, A., Singh, A. P., and Raman, B. (2026). EarthESND: Lightweight multiscale echo state network with dendritic neural network readout for earthquake early warning. *Computers and Electrical Engineering*. (Reference implementation documented in docs/PAPER_REPRODUCTION.md; not the primary geographic objective of this programme.)

Krischer, L., Megies, T., Barsch, R., Beyreuther, M., Lecocq, T., Caudron, C., and Wassermann, J. (2015). ObsPy: A bridge for earthquake science. *Seismological Research Letters*, 86(3), 765–771. <https://doi.org/10.1785/0220150032>

International Federation of Digital Seismograph Networks (FDSN). (n.d.). FDSN web services specification. <https://www.fdsn.org/webservices/> (accessed 29 July 2026).

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ORFEUS Data Center / EIDA. (n.d.). European integrated waveform archives. <https://www.orfeus-eu.org/> (accessed 29 July 2026).

Project documentation: docs/PROJECT_CHARTER.md; docs/RESEARCH_DIRECTION.md; docs/IRIS_DATASET_REPORT.md; docs/EDA_REPORT.md; docs/SIGNAL_ANALYSIS_REPORT.md; docs/PREPROCESSING_REPORT.md; docs/FEATURE_ENGINEERING_REPORT.md; docs/RESULTS_AND_DISCUSSION.md; docs/REPRODUCIBILITY_STATEMENT.md; docs/DATA_AVAILABILITY_STATEMENT.md; docs/FINAL_DELIVERABLE_AUDIT.md.

15. Statement of evidence support

Statement class	Support
Numeric results (SNR, picks, magnitudes, feature means)	Supported — JSON/CSV cited in phase reports and summarised in §8
Figure content	Supported — PNG paths verified in docs/FINAL_DELIVERABLE_AUDIT.md (24/24 local)
ML model performance	Not claimed — no training artefacts in repository
Operational EEW deployment	Not claimed — explicitly out of scope
European dataset results	Not claimed — proposed only
Author name / CERG edition	Not in repository — placeholder avoided; programme name from charter only
EarthESND 88 tests	Supported — docs/PROJECT_CHARTER.md, README.md
Git clone contains all PNG/MiniSEED	Qualified — audit WARNING; see Appendix C

Appendix A — Pilot event and station tables

Table A1. Event summary (full CSV: reports/tables/event_summary.csv)

short_id	Magnitude	Type	Station	Origin (UTC)
ci40964384	4.54	ml	CI.ADO	2024-10-25 15:04:23
ci40964128	4.72	mw	CI.ADO	2024-10-25 08:05:08
ci40964048	4.16	mw	CI.ADO	2024-10-25 06:52:08
nn00882322	4.39	ml	CI.ADO	2024-08-14 03:06:35
ci40699207	4.39	mw	CI.USC	2024-08-12 19:20:24
ci40675215	4.87	mw	CI.ADO	2024-07-29 20:00:52
ci40593503	4.05	mw	CI.ADO	2024-05-20 12:17:36
ci40735352	4.11	mw	CI.USC	2024-05-01 20:49:00

Table A2. Station summary (reports/tables/station_summary.csv)

Station	Events	Mean fs (Hz)	Duration (s)
CI.ADO.–.BHZ	6	40.0	300.0
CI.USC.–.BHZ	2	40.0	300.0

Appendix B — Reproducibility statement (D-S1)

The following text is the repository reproducibility statement (docs/REPRODUCIBILITY_STATEMENT.md), included for submission. Figure paths in that file use repository-root notation; PNG outputs appear under reports/figures/ relative to the proposed work.

B.1 What this workflow reproduces

The pilot chain in scripts/download/ and scripts/analysis/ produces: manifest and MiniSEED (when downloaded); EDA summaries and four figures; signal metrics, eight figures, and CSV; preprocessing NPZ, eight figures, QC table; 8×18 feature matrix and four feature figures. Interpretation is in phase docs and prior proposals. California results did not use EarthESND src/ modules.

B.2 Software environment

Python ≥ 3.11; ObsPy, NumPy, Pandas, Matplotlib, SciPy, PyYAML per pyproject.toml. Suggested install: `pip install -e ".[dev]"` from repository root. No separate requirements.txt.

B.3 Execution order

```
python scripts/download/download_iris_california_pilot.py --event-count 8 --m
in-events 5
```

```
python scripts/analysis/run_california_pilot_eda.py
```

```
python scripts/analysis/run_california_signal_analysis.py
```

```
python scripts/analysis/run_california_preprocessing.py
```

```
python scripts/analysis/run_california_feature_engineering.py
```

Defaults: data/raw/iris/, data/manifests/iris_california_pilot_events.csv, logs/iris_california_pilot_summary.json. Preprocessing config: reports/preprocessing/preprocessing_config.json.

B.4 Included vs excluded artefacts

Typically in Git: scripts, manifest, phase docs, reports/tables/, JSON summaries, NPZ, feature matrix (see audit).

Gitignored: data/raw/, logs/, figures/ (includes reports/figures/). Fresh clones must regenerate or use a separate archive (docs/FINAL_DELIVERABLE_AUDIT.md).

B.5 Reproducibility limitations

FDSN rerun variability; fixed station list; origin-aligned windows; 0/8 response removal; N=8; EarthESND not on pilot path; PNGs may differ slightly by library version.

B.6 Verification

Compare to reports/eda/dataset_summary.json,
reports/signal_analysis/signal_analysis_summary.json,
reports/preprocessing/preprocessing_summary.json,
reports/features/feature_summary.json.

Appendix C — Data availability and ethics statement (D-S2)

Condensed from docs/DATA_AVAILABILITY_STATEMENT.md for this submission package.

C.1 Data sources

Role	Provider	Access
Catalog	USGS FDSN Event	Public — https://earthquake.usgs.gov
Waveforms	EarthScope	Public FDSN — https://service.earthscope.org

Query details: docs/IRIS_DATASET_REPORT.md. Eight events, magnitudes 4.05–4.87, CI.ADO (6) / CI.USC (2). No credentials required.

C.2 Repository availability

In Git (typical): manifest, derived reports/ products, scripts, documentation.

Not in Git: raw MiniSEED (data/raw/), logs (logs/), figures (reports/figures/ per .gitignore rule figures/).

Regeneration: download script + analysis pipeline (Appendix B).

C.3 Ethics

Public seismic data only; no human subjects; cite USGS, EarthScope, and FDSN terms. GDPR note applies to future European integrations, not the completed California pilot.

C.4 Data-use limitations

BHZ only; N=8; ADO bias; legacy files outside manifest; counts not physical motion; not EarthESND geography.

Full citations: see docs/DATA_AVAILABILITY_STATEMENT.md §7 and Section 14 of this document.

Appendix D — Supplementary per-event figures

Remaining signal analysis panels (four-panel: raw, normalised, STA/LTA, P marker):

Fig.	File
D1	figures/signal_analysis_ci40593503.png
D2	figures/signal_analysis_ci40735352.png
D3	figures/signal_analysis_ci40964048.png
D4	figures/signal_analysis_ci40964128.png
D5	figures/signal_analysis_ci40964384.png
D6	figures/signal_analysis_nn00882322.png

Remaining preprocessing panels (raw → detrended → filtered → normalised):

Fig.	File
D7	figures/preprocessing_ci40593503.png
D8	figures/preprocessing_ci40735352.png
D9	figures/preprocessing_ci40964048.png
D10	figures/preprocessing_ci40964128.png
D11	figures/preprocessing_ci40964384.png
D12	figures/preprocessing_nn00882322.png

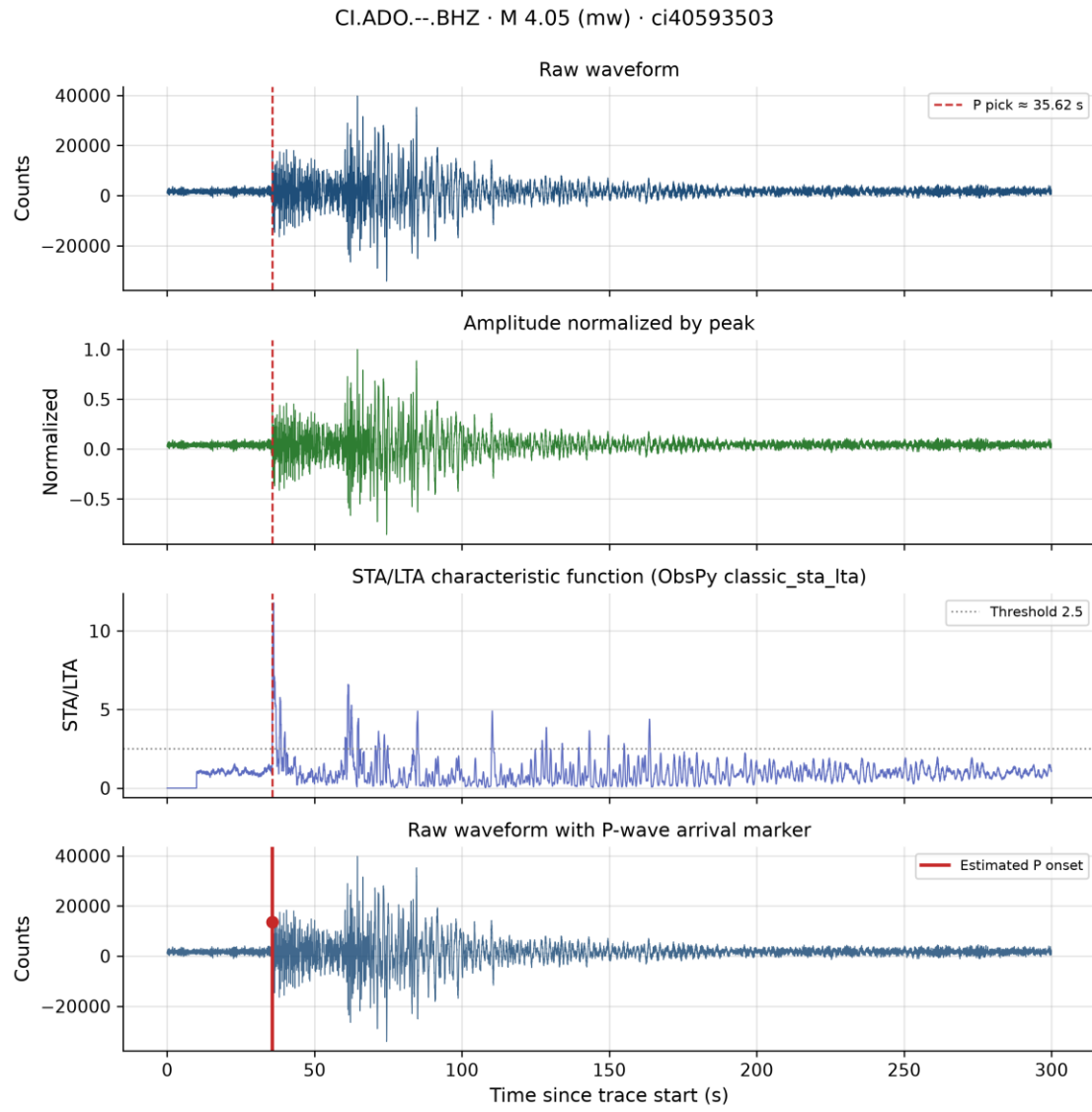


Figure D1: Signal analysis — ci40593503.

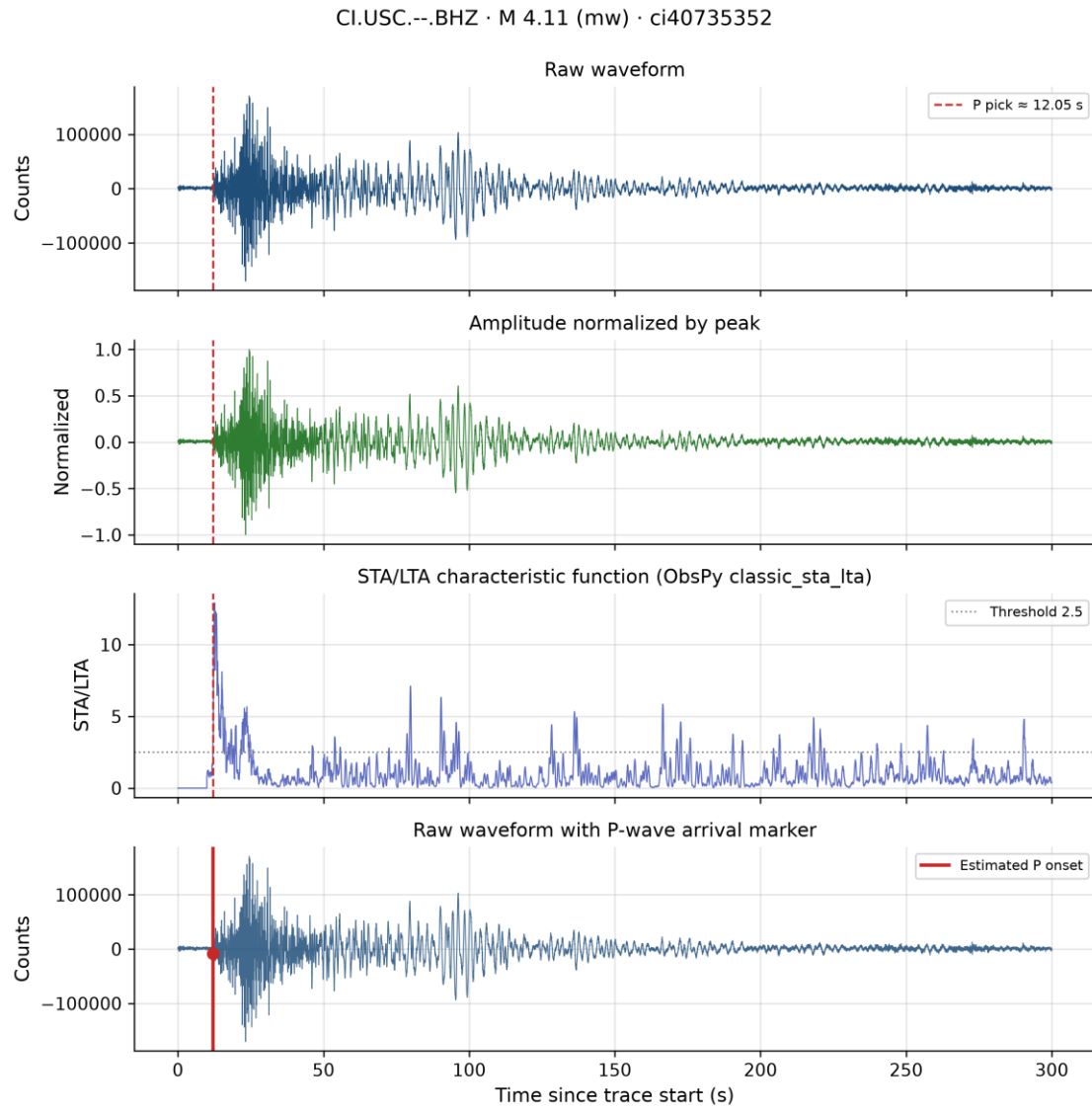


Figure D2: Signal analysis — ci40735352.

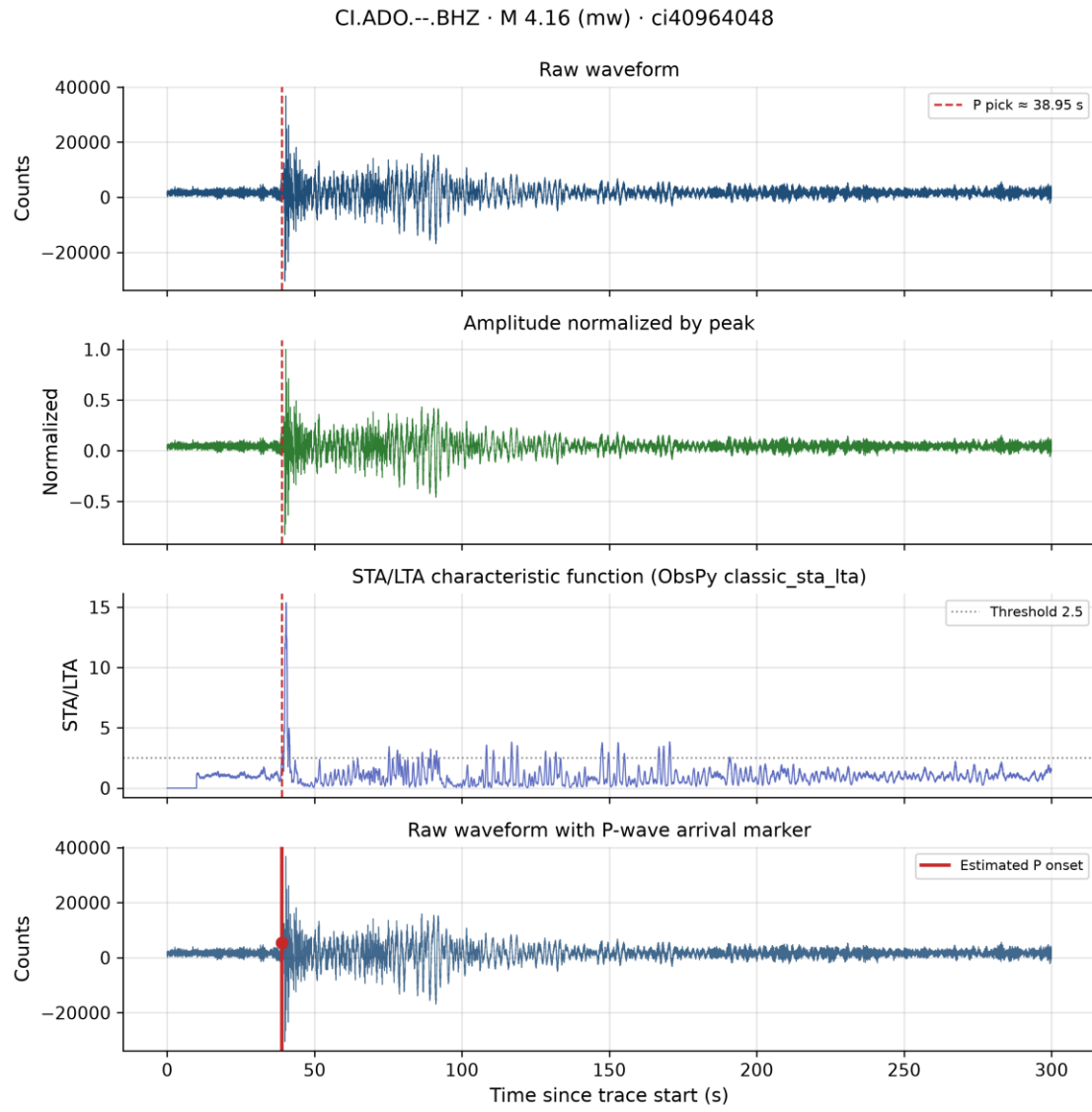


Figure D3: Signal analysis — ci40964048.

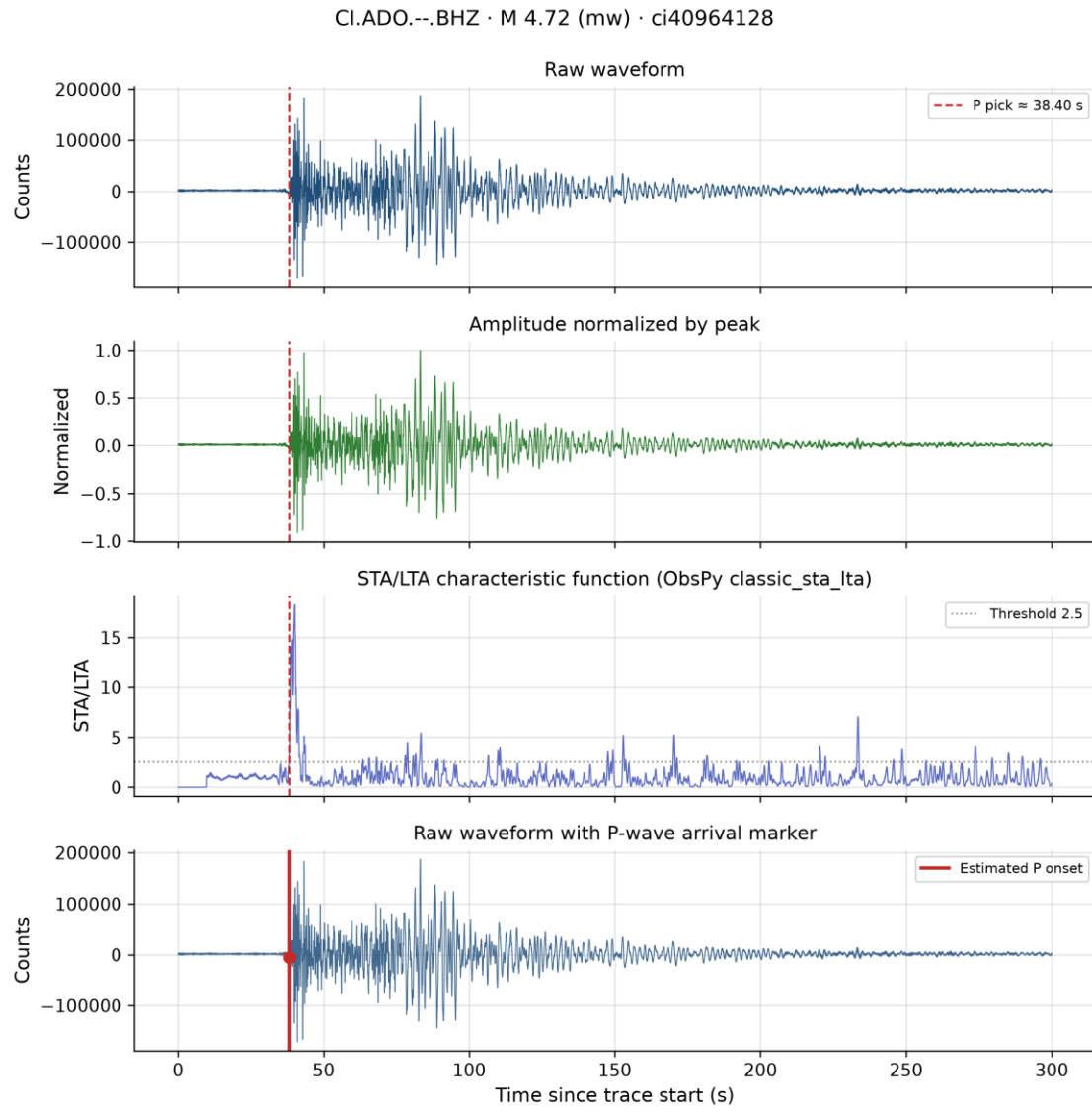


Figure D4: Signal analysis — ci40964128.

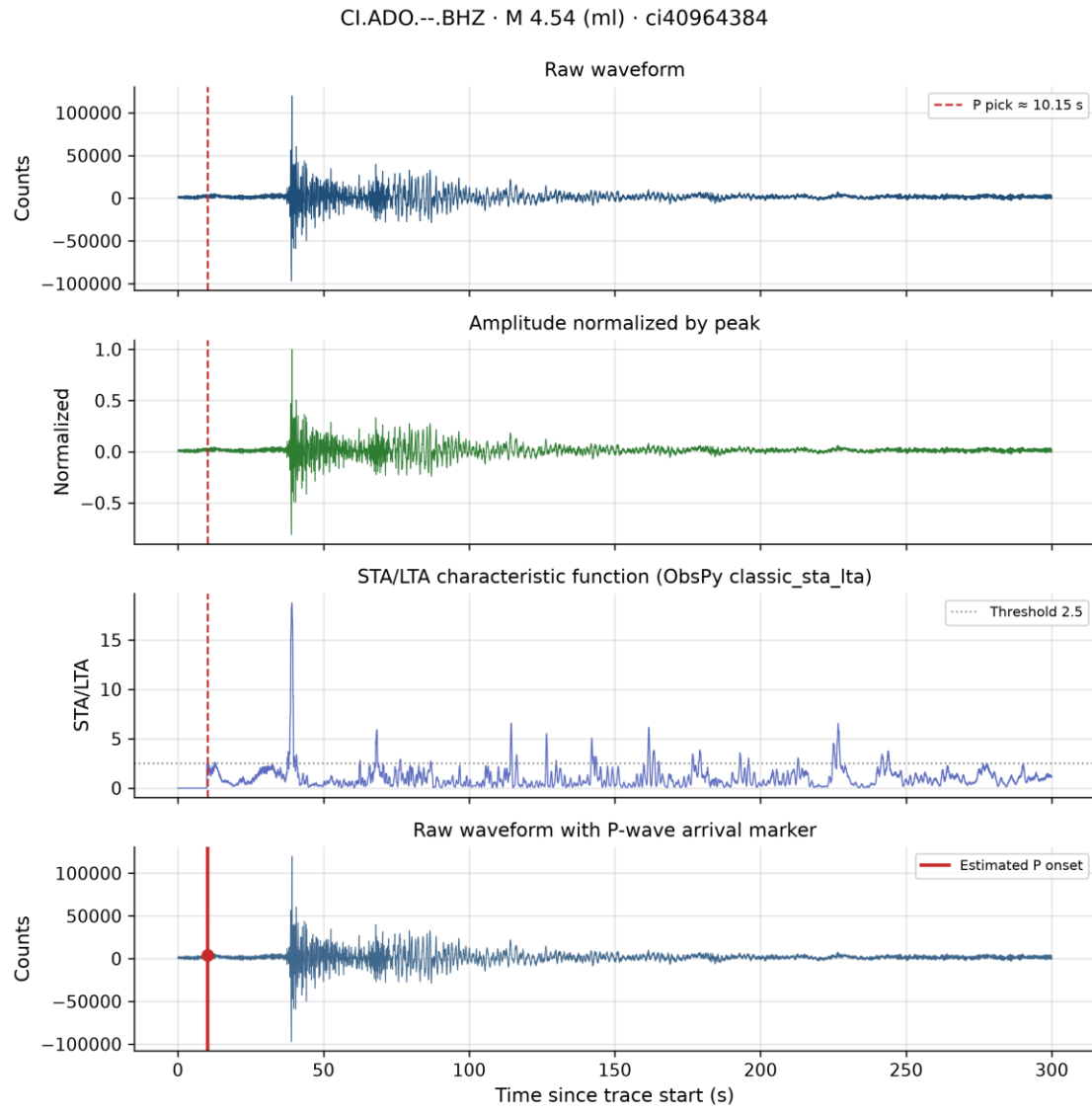


Figure D5: Signal analysis — ci40964384.

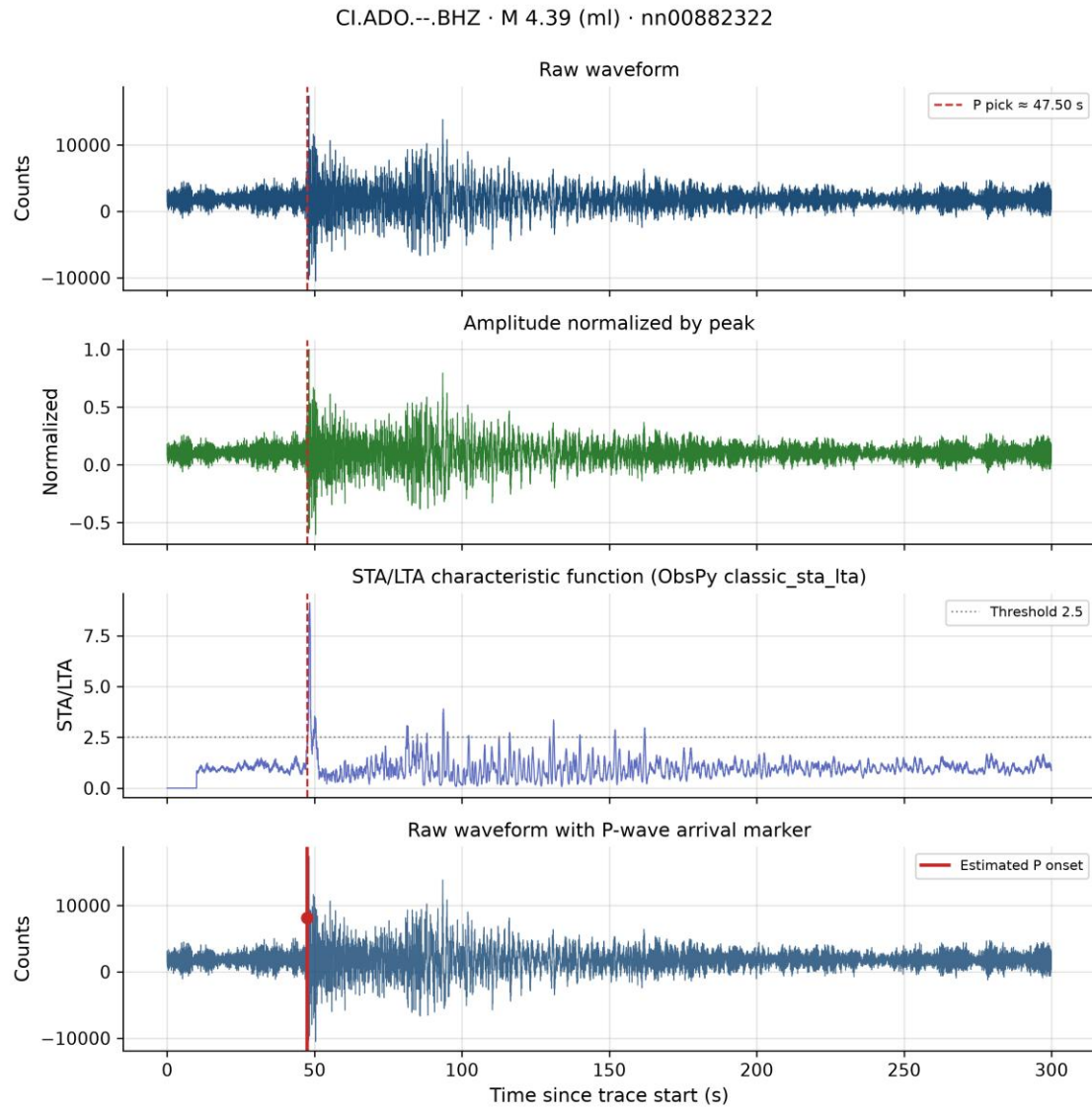


Figure D6: Signal analysis — nn00882322.

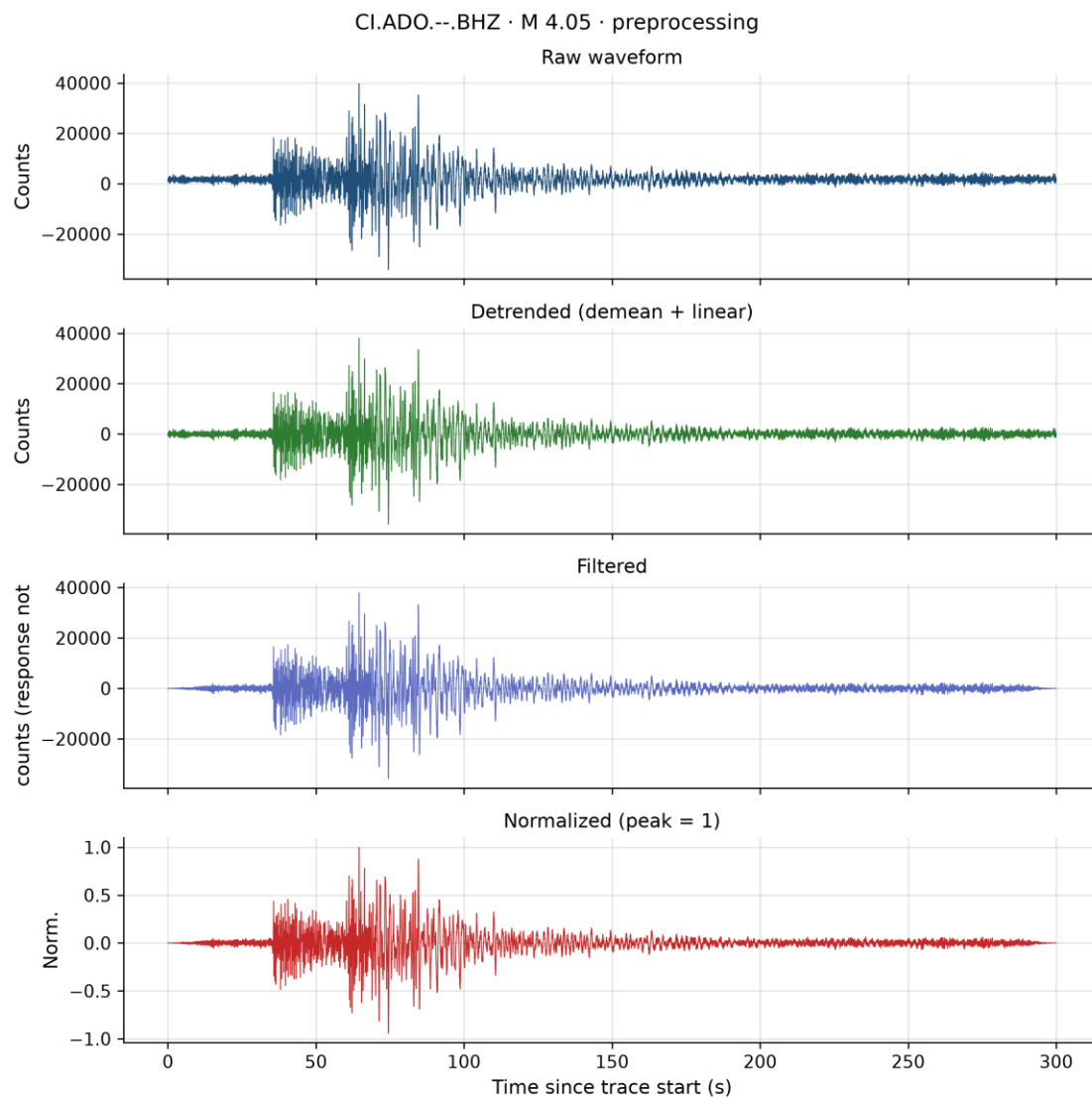


Figure D7: Preprocessing — ci40593503.

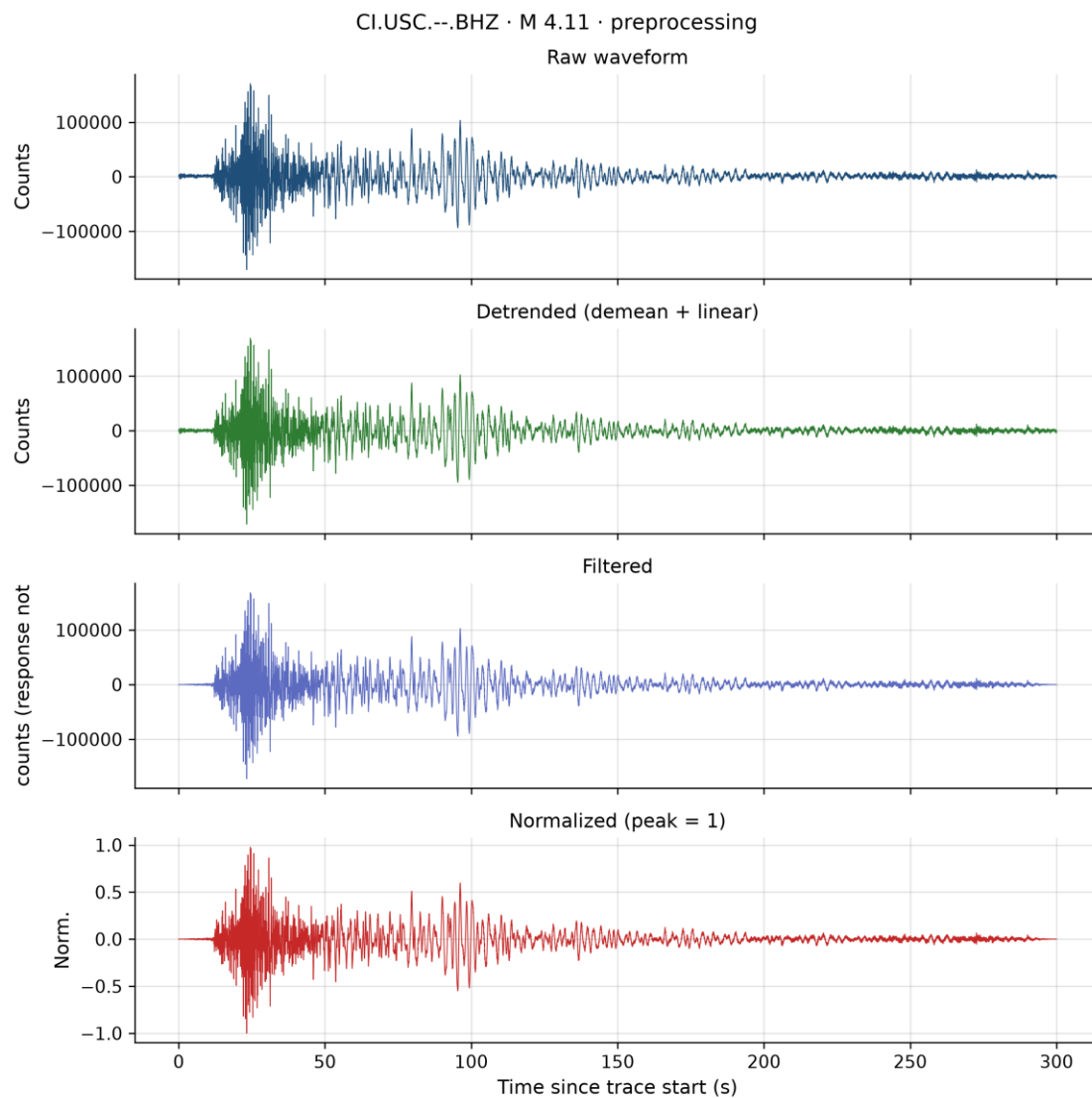


Figure D8: Preprocessing — ci40735352.

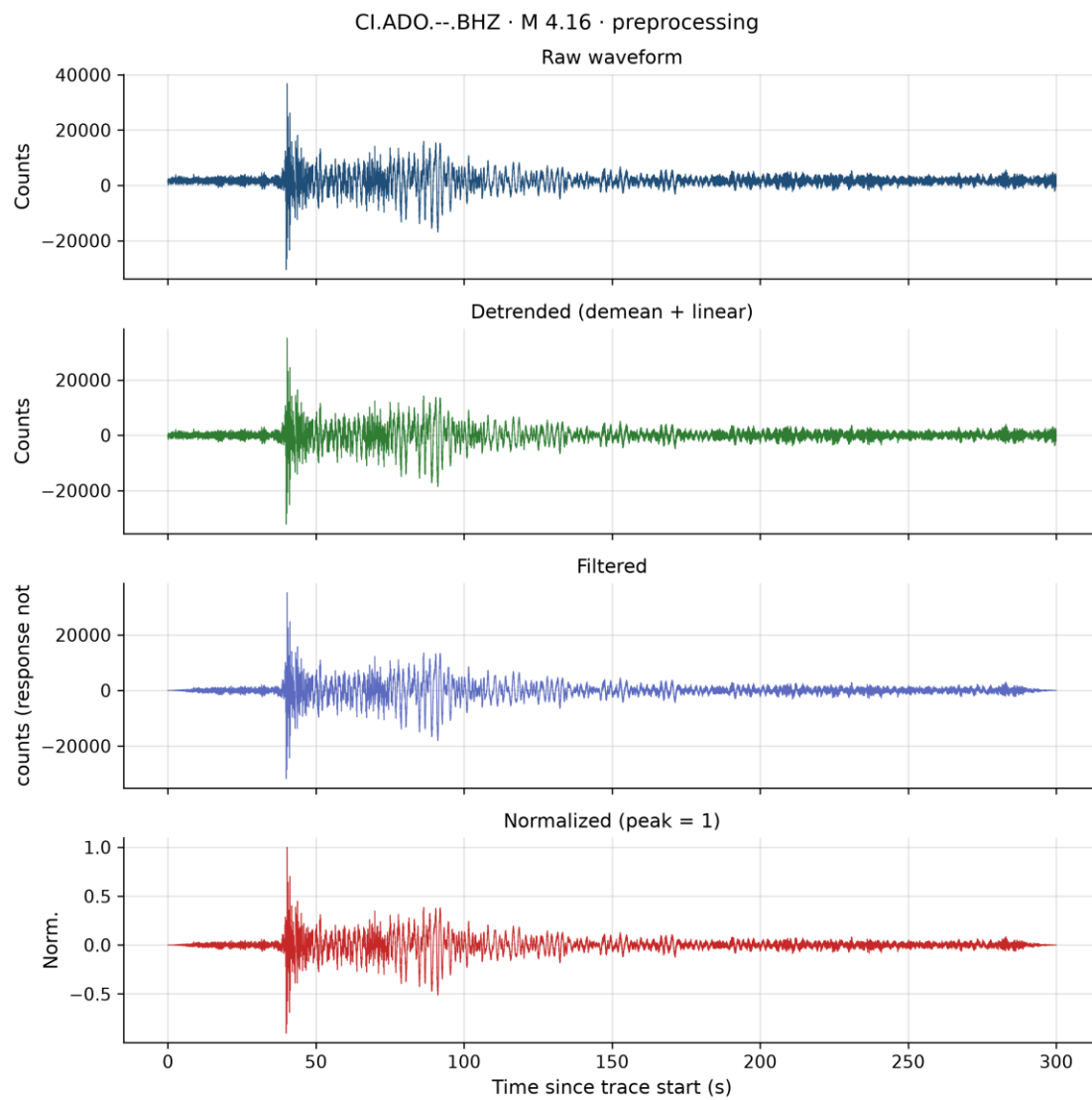


Figure D9: Preprocessing — ci40964048.

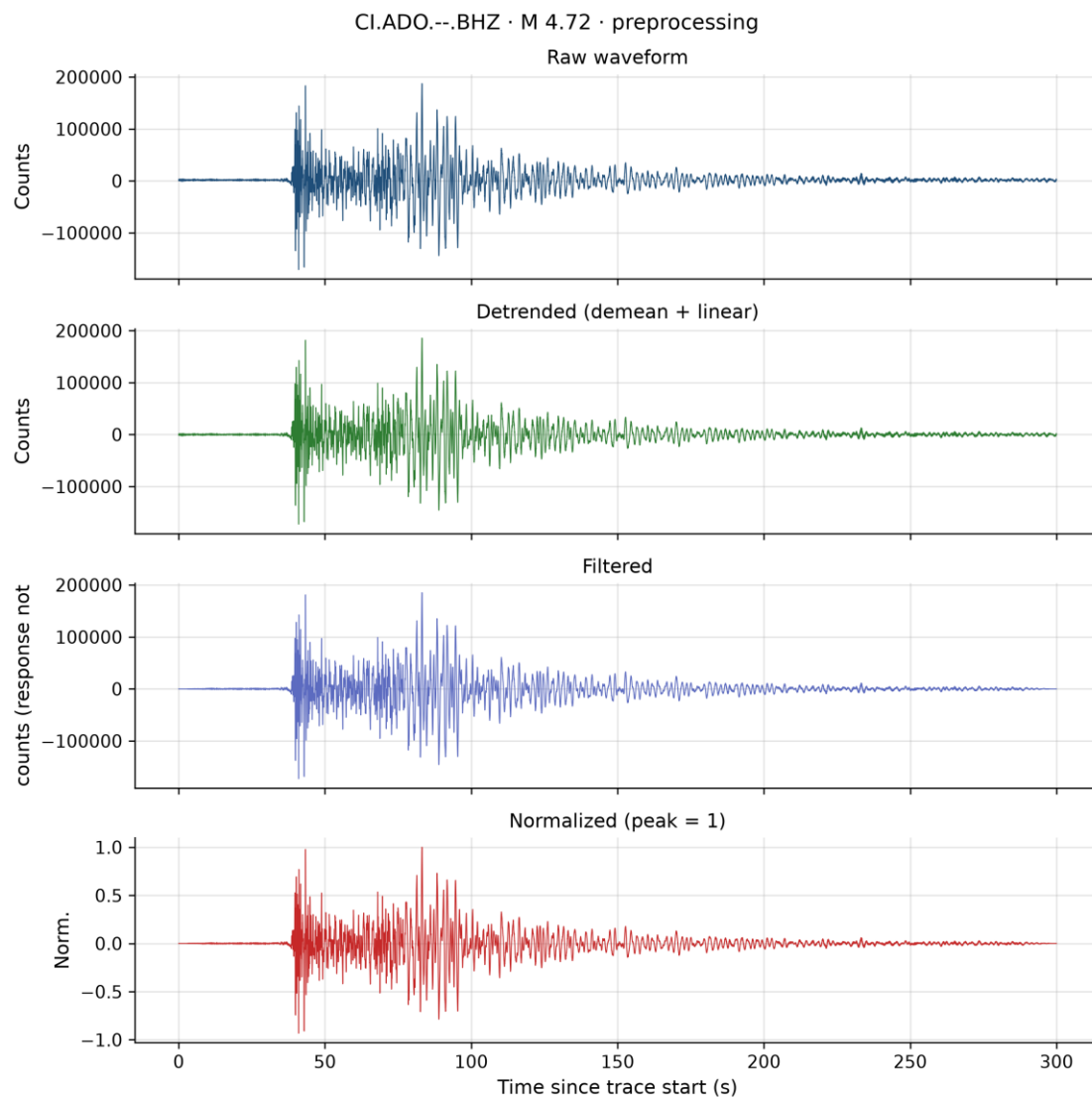


Figure D10: Preprocessing — ci40964128.

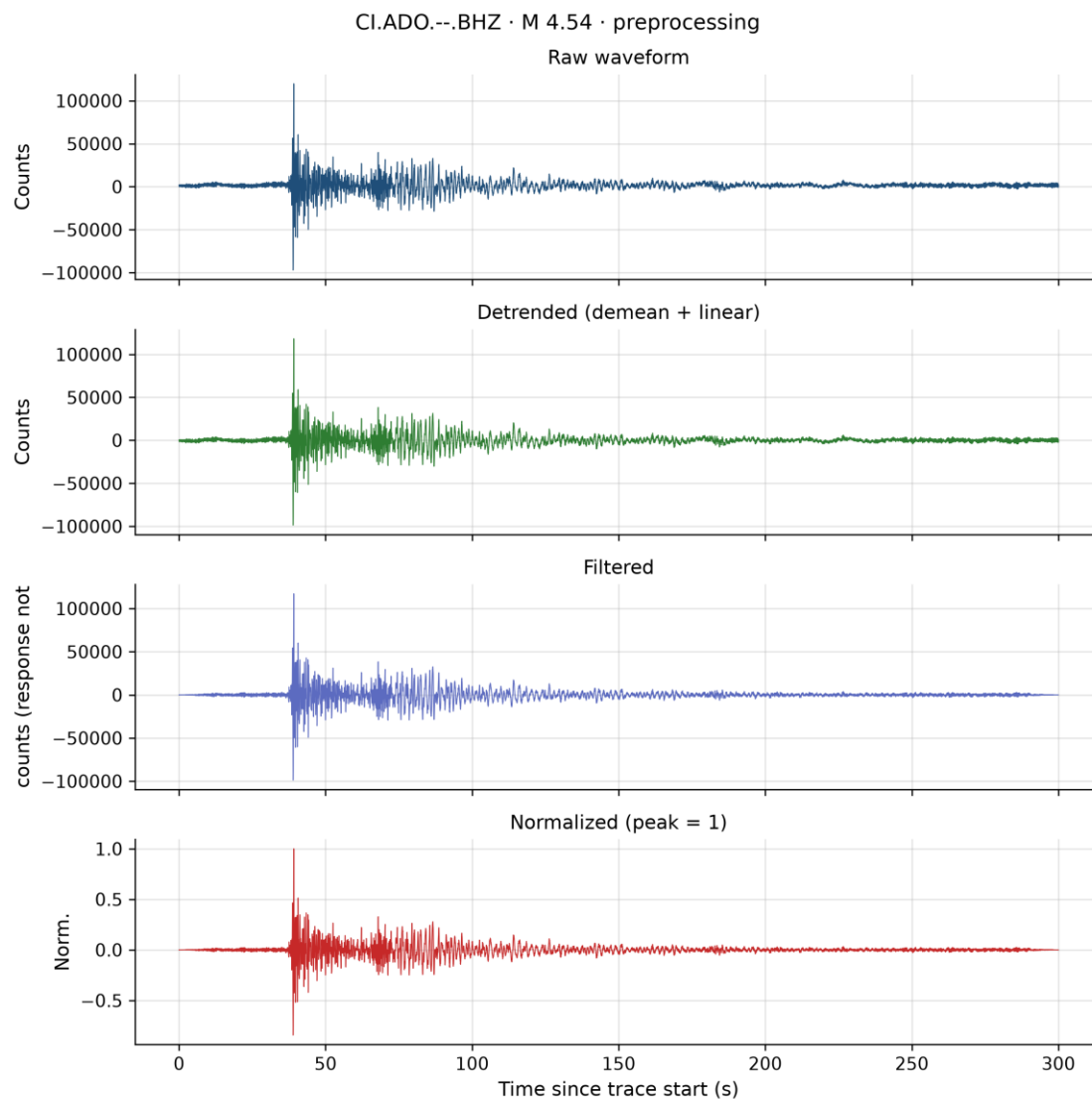


Figure D11: Preprocessing — ci40964384.

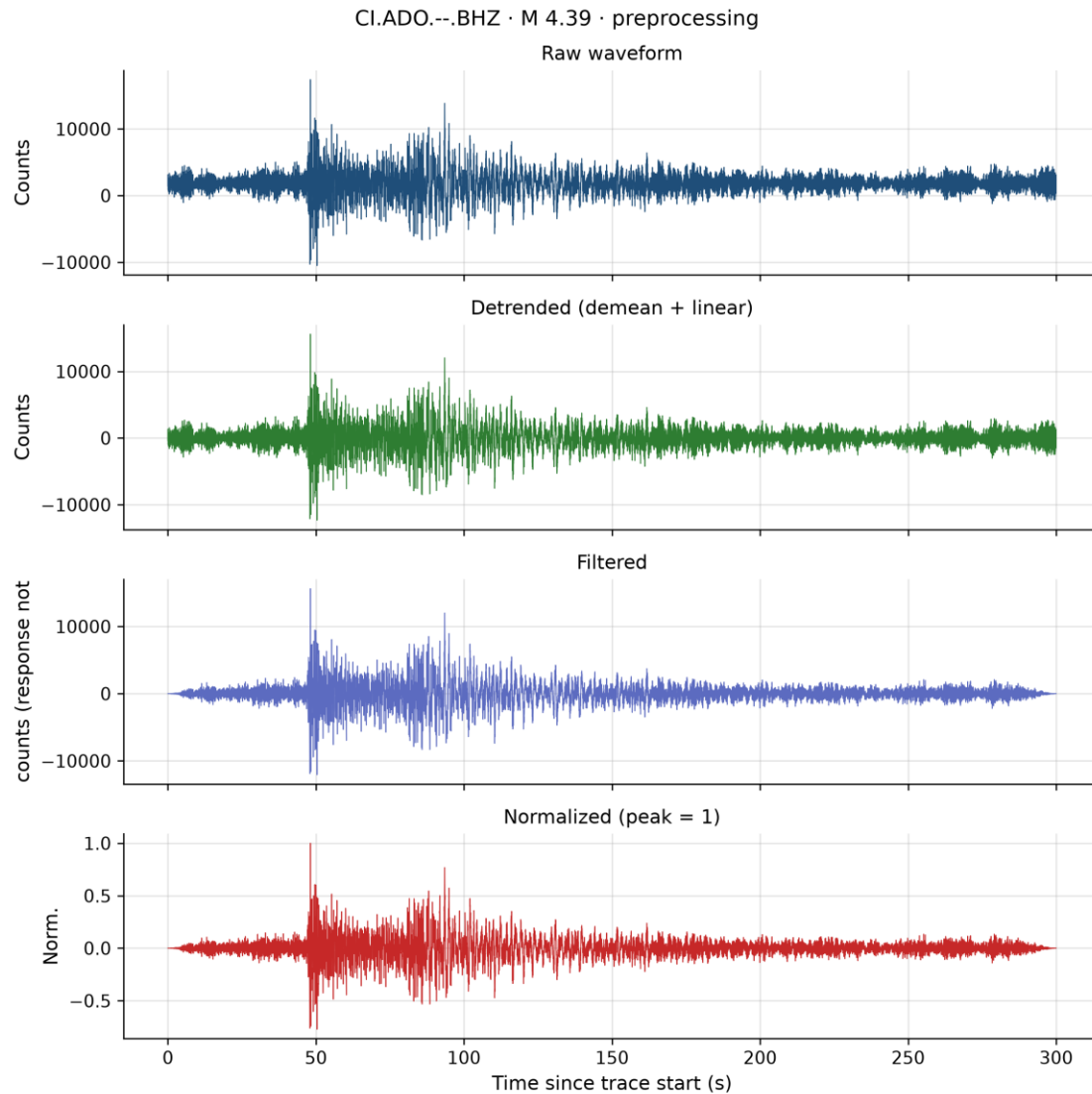


Figure D12: Preprocessing — nn00882322.

Appendix E — Machine-readable summary index

File	Content
reports/eda/dataset_summary.json	EDA cohort statistics
reports/signal_analysis/signal_analysis_summary.json	Signal aggregates
reports/preprocessing/preprocessing_summary.json	Preprocessing aggregates
reports/features/feature_summary.json	Feature cohort statistics

File	Content
reports/preprocessing/preprocessing_config.json	Frozen B.1 parameters

Appendix F — Programme positioning and document scope (supplementary)

The following material supports Sections 1–6 but is placed in supplementary appendix per review guidance.

F.1 Programme positioning

The project was initiated within a Swiss certificate programme in geophysical risk assessment, aligned with European expectations for traceable analysis (open data, documented processing, explicit limitations). Initial effort targeted reproduction of the EarthSND architecture (Joshi, Singh, and Raman, 2026), producing a test-backed reference codebase preserved for methodological comparison only. Programme guidance subsequently reframed the work toward a logical European Earthquake Early Warning research project with real waveforms, theory, visualisation, tables, and formal reporting, using California FDSN data as a completed pilot laboratory (docs/PROJECT_CHARTER.md, docs/RESEARCH_DIRECTION.md).

F.2 Scope of the proposed work

The final deliverable (D-F1) supersedes reports/Research_Proposal_v1.md as the submission-oriented consolidation. It synthesises completed pilot work (acquisition through interpretation) and proposed extensions (European data, scale, artificial intelligence evaluation, real-time paths). Sources: phase reports under docs/, artefacts under reports/, docs/REPRODUCIBILITY_STATEMENT.md, docs/DATA_AVAILABILITY_STATEMENT.md, and docs/FINAL_DELIVERABLE_AUDIT.md.

F.3 Document control

Field	Value
Deliverable	D-F1 Final Research Proposal / Technical Report
Version	1.0
Date	29 July 2026
Prior version	Research Proposal v1.0 (reports/Research_Proposal_v1.md)
Pilot freeze date	2026-07-29 (phase reports)